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LIGHT-EMITTING DEVICE INCLUDING HETEROCYCLIC COMPOUND, ELECTRONIC APPARATUS AND ELECTRONIC EQUIPMENT INCLUDING THE LIGHT-EMITTING DEVICE, AND THE HETEROCYCLIC COMPOUND

Abstract

Embodiments provide a heterocyclic compound, a light-emitting device including the heterocyclic compound, an electronic apparatus including the light-emitting device, and an electronic equipment including the light-emitting device. The heterocyclic compound includes an electron donor moiety including nitrogen, an electron acceptor moiety including boron, and a spiro core atom, wherein the spiro core atom is bonded to the electron donor moiety via a first bond and a second bond, and the spiro core atom is bonded to the electron acceptor moiety via a third bond and a fourth bond. The first bond, the second bond, the third bond, and the fourth bond are each a single bond, and the heterocyclic compound satisfies at least one of Conditions 1 to 3, which are explained in the specification.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
[0001] This application claims priority to and benefits of Korean Patent Application No. 10-2024-0028156 under 35 U.S.C. § 119, filed on Feb. 27, 2024, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.
BACKGROUND
1. Technical Field
[0002] Embodiments relate to a light-emitting device including a heterocyclic compound, an electronic apparatus and electronic equipment that include the light-emitting device, and the heterocyclic compound.
2. Description of the Related Art
[0003] Light-emitting devices are self-emissive devices that have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of luminance, driving voltage, and response speed.
[0004] In a light-emitting device, a first electrode may be arranged on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode may be sequentially arranged on the first electrode. Holes provided from the first electrode move toward the emission layer through the hole transport region, and electrons provided from the second electrode move toward the emission layer through the electron transport region. Carriers, such as the holes and electrons, recombine in the emission layer to produce excitons. The excitons transition from an excited state to a ground state to thereby generate light.
[0005] It is to be understood that this background of the technology section is, in part, intended to provide useful background for understanding the technology. However, this background of the technology section may also include ideas, concepts, or recognitions that were not part of what was known or appreciated by those skilled in the pertinent art prior to a corresponding effective filing date of the subject matter disclosed herein.
SUMMARY
[0006] Embodiments include a light-emitting device including a heterocyclic compound, an electronic apparatus and electronic equipment that include the light-emitting device, and the heterocyclic compound.
[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the embodiments of the disclosure.
[0008] According to embodiments, a light-emitting device may include [0009] a first electrode,

[0010] a second electrode facing the first electrode, [0011] an interlayer between the first electrode and the second electrode and including an emission layer, and [0012] a heterocyclic compound including an electron donor moiety including nitrogen, an electron acceptor moiety including boron, and a spiro core atom, wherein [0013] the spiro core atom may be bonded to the electron donor moiety via a first bond and a second bond, [0014] the spiro core atom may be bonded to the electron acceptor moiety via a third bond and a fourth bond, [0015] the first bond, the second bond, the third bond, and the fourth bond may each be a single bond, and [0016] at least one of Conditions 1 to 3 may be satisfied:

[Condition 1]

[0017] the electron donor moiety includes a group represented by Formula 1A;

[Condition 2]

[0018] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the first ring and the second ring are linked to each other via a first linking group;

[Condition 3]

[0019] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the second ring and the third ring are linked to each other via a second linking group;

##STR00001##

[0020] In Formula 1A, [0021] R.sub.61 to R.sub.63 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), [0022] a61 and a63 may each independently be an integer from 1 to 5, [0023] a62 may be an integer from 1 to 3, [0024] * indicates a binding site to a neighboring atom, [0025] R.sub.10a may be:

[0026] hydrogen, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

[0027] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or any combination thereof;

[0028] a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or any combination thereof;

or
[0029] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32), and
[0030] Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
[0031] In an embodiment, the emission layer may include a host and a dopant, and the dopant may include the heterocyclic compound.
[0032] In an embodiment, the light-emitting device may further include: a second compound including at least one π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group; or a third compound may include a group represented by Formula 3; or a fourth compound that may emit delayed fluorescence; or any combination thereof, wherein the heterocyclic compound, the second compound, the third compound, and the fourth compound may be different from each other, and wherein Formula 3 is explained below.
[0033] In an embodiment, the second compound may include a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or any combination thereof; and the fourth compound may be a compound including at least one cyclic group including both boron (B) and nitrogen (N) as ring-forming atoms.
[0034] In an embodiment, the emission layer may include: the heterocyclic compound; and the second compound, the third compound, the fourth compound, or any combination thereof; and the emission layer may emit blue light.
[0035] According to embodiments, an electronic apparatus may include the light-emitting device.
[0036] In an embodiment, the electronic apparatus may further include a thin-film transistor, wherein the thin-film transistor may include a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically connected to at least one of the source electrode and the drain electrode.
[0037] In an embodiment, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof.
[0038] According to embodiments, an electronic equipment may include the light-emitting device.
[0039] In an embodiment, the electronic equipment may be a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.
[0040] According to embodiments, a heterocyclic compound may include an electron donor moiety including nitrogen, an electron acceptor moiety including boron, and a spiro core atom, wherein
[0041] the spiro core atom may be bonded to the electron donor moiety via a first bond and a second bond; the spiro core atom may be bonded to the electron acceptor moiety via a third bond and a fourth bond; the first bond, the second bond, the third bond, and the fourth bond may each be a single bond; and at least one of Conditions 1 to 3 may be satisfied, which are explained herein.
[0042] In an embodiment, the heterocyclic compound may be represented by Formula 1:
##STR00002##

[0043] In Formula 1, [0044] T.sub.1 may be C, Si, or Ge, [0045] ring CY.sub.1 to ring CY.sub.5 may each independently be a C.sub.5-C.sub.30 carbocyclic group or a C.sub.1-C.sub.30 heterocyclic group, [0046] T.sub.21 may be *—C(Z.sub.1)(Z.sub.2)—*, *—Si(Z.sub.1)(Z.sub.2)—**, *—N(Z.sub.1)—**, *—O—**, or *—S—*, [0047] T.sub.22 may be *—C(Z.sub.3)(Z.sub.4)—**, *—Si(Z.sub.3)(Z.sub.4)—**, *—N(Z.sub.3)—**, *—O—**, or *—S—**, [0048] R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.5-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), [0049] a1 to a5 may each independently be an integer from 1 to 20, [0050] Ar.sub.1 may be a group represented by Formula 1A, a C.sub.5-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0051] Z.sub.1 and Z.sub.2 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0052] Z.sub.3 and Z.sub.4 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0053] two or more neighboring groups among R.sub.1 in the number of a1, R.sub.2 in the number of a2, R.sub.3 in the number of a3, R.sub.4 in the number of a4, R.sub.5 in the number of a5, Z.sub.1 to Z.sub.4, and Ar.sub.1 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and [0054] R.sub.10a and Q.sub.1 to Q.sub.3 may each be the same as described herein.

[0055] In an embodiment, ring CY.sub.1 to ring CY.sub.5 may each independently be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an azafluorene group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a dibenzooxasiline group, a dibenzothiasiline group, a dibenzodihydroazasiline group, a dibenzodihydrodihydrodisiline group, a dibenzodihydrosiline group, a dibenzodioxine group, a dibenzooxathiine group, a dibenzooxazine group, a dibenzopyran group, a dibenzodithiine group, a dibenzothiazine group, a dibenzothiopyran group, a dibenzocyclohexadiene group, a dibenzodihydropyridine group, or a dibenzodihydropyrazine group.

[0056] In an embodiment, T.sub.21 may be *—C(Z.sub.1)(Z.sub.2)—** or *—Si(Z.sub.1)(Z.sub.2)—**; and T.sub.22 may be *—C(Z.sub.3)(Z.sub.4)—** or *—Si(Z.sub.3)(Z.sub.4)—**.

[0057] In an embodiment, at least one of Conditions 4 and 5 may be satisfied, which are explained below.

[0058] In an embodiment, R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 may each independently be: [0059] hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; [0060] a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof; [0061] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, or an azadibenzosilolyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof; or [0062] —C(Q.sub.1)(Q.sub.2)

(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), and [0063] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
[0064] In an embodiment, Ar.sub.1 may be: [0065] a group represented by Formula 1A; or [0066] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, or an azadibenzosilolyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof, and [0067] Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

[0068] In an embodiment, Ar.sub.1 may be a group represented by Formula 1A.
[0069] In an embodiment, T.sub.21 and T.sub.22 may each independently be a group represented by Formula 1B, which is explained below.
[0070] In an embodiment, the heterocyclic compound may be represented by Formula 1-1, which is explained below.
[0071] It is to be understood that the embodiments above are described in a generic and explanatory sense only and not for the purposes of limitation, and the disclosure is not limited to the embodiments described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0072] The accompanying drawings are included to provide a further understanding of the embodiments, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and principles thereof. The above and other aspects and features of the disclosure will become more apparent by describing in detail embodiments thereof with reference to the accompanying drawings, in which:
[0073] FIG. **1** is a schematic cross-sectional view of a light-emitting device according to an embodiment;
[0074] FIG. **2** is a schematic cross-sectional view of an electronic apparatus according to an embodiment;
[0075] FIG. **3** is a schematic cross-sectional view of an electronic apparatus according to another embodiment; and
[0076] FIG. **4** is a schematic perspective view of an electronic equipment according to an embodiment;
[0077] FIG. **5** is a schematic perspective view of an exterior of a vehicle as an electronic equipment including a light-emitting device according to an embodiment; and
[0078] FIGS. **6**A to **6**C are each a schematic diagram of an interior of a vehicle according to embodiments.
DETAILED DESCRIPTION OF THE EMBODIMENTS
[0079] The disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments are shown. This disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art.
[0080] In the drawings, the sizes, thicknesses, ratios, and dimensions of the elements may be exaggerated for ease of description and for clarity. Like reference numbers and/or like reference characters refer to like elements throughout.
[0081] In the description, it will be understood that when an element (or region, layer, part, etc.) is referred to as being “on”, “connected to”, or “coupled to” another element, it can be directly on, connected to, or coupled to the other element, or one or more intervening elements may be present therebetween. In a similar sense, when an element (or region, layer, part, etc.) is described as “covering” another element, it can directly cover the other element, or one or more intervening elements may be present therebetween.
[0082] In the description, when an element is “directly on,” “directly connected to,” or “directly coupled to” another element, there are no intervening elements present. For example, “directly on” may mean that two layers or two elements are disposed without an additional element such as an adhesion element therebetween.
[0083] As used herein, the expressions used in the singular such as “a,” “an,” and “the,” are

intended to include the plural forms as well, unless the context clearly indicates otherwise.
[0084] As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. For example, “A and/or B” may be understood to mean “A, B, or A and B.” The terms “and” and “or” may be used in the conjunctive or disjunctive sense and may be understood to be equivalent to “and/or”.
[0085] In the specification and the claims, the term “at least one of” is intended to include the meaning of “at least one selected from the group consisting of” for the purpose of its meaning and interpretation. For example, “at least one of A, B, and C” may be understood to mean A only, B only, C only, or any combination of two or more of A, B, and C, such as ABC, ACC, BC, or CC. When preceding a list of elements, the term, “at least one of,” modifies the entire list of elements and does not modify the individual elements of the list.
[0086] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element could be termed a second element without departing from the teachings of the disclosure. Similarly, a second element could be termed a first element, without departing from the scope of the disclosure.
[0087] The spatially relative terms “below”, “beneath”, “lower”, “above”, “upper”, or the like, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device illustrated in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in other directions and thus the spatially relative terms may be interpreted differently depending on the orientations.
[0088] The terms “about” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the recited value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the recited quantity (for example, the limitations of the measurement system). For example, “about” may mean within one or more standard deviations, or within +20%, +10%, or +5% of the stated value.
[0089] It should be understood that the terms “comprises,” “comprising,” “includes,” “including,” “have,” “having,” “contains,” “containing,” and the like are intended to specify the presence of stated features, integers, steps, operations, elements, components, or combinations thereof in the disclosure, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.
[0090] Unless otherwise defined or implied herein, all terms (including technical and scientific terms) used have the same meaning as commonly understood by those skilled in the art to which this disclosure pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an ideal or excessively formal sense unless clearly defined in the specification.
[0091] According to an embodiment, a light-emitting device may include: [0092] a first electrode; [0093] a second electrode facing the first electrode; [0094] an interlayer between the first electrode and the second electrode and including an emission layer; and [0095] a heterocyclic compound including: an electron donor moiety including nitrogen; [0096] an electron acceptor moiety including boron; and a spiro core atom, wherein [0097] the spiro core atom may be bonded to the electron donor moiety via a first bond and a second bond, [0098] the spiro core atom may be bonded to the electron acceptor moiety via a third bond and a fourth bond, [0099] the first bond, the second bond, the third bond, and the fourth bond may each be a single bond, and at least one of

Conditions 1 to 3 may be satisfied:
[Condition 1]
[0100] the electron donor moiety includes a group represented by Formula 1A;
[Condition 2]
[0101] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the first ring and the second ring are linked to each other via a first linking group;
[Condition 3]
[0102] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the second ring and the third ring are linked to each other via a second linking group;
##STR00003##
[0103] In Formula 1A, [0104] R.sub.61 to R.sub.63 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), [0105] a61 and a63 may each independently be an integer from 1 to 5, [0106] a62 may be an integer from 1 to 3, [0107] * indicates a binding site to a neighboring atom, and [0108] R.sub.10a and Q.sub.1 to Q.sub.3 may each independently be the same as described herein.
[0109] In an embodiment, [0110] the first electrode may be an anode, [0111] the second electrode may be a cathode, [0112] the interlayer may further include a hole transport region between the first electrode and the emission layer, and an electron transport region between the emission layer and the second electrode, [0113] the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and [0114] the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.
[0115] In embodiments, the interlayer may include the heterocyclic compound.
[0116] In embodiments, the emission layer may include the heterocyclic compound.
[0117] In embodiments, the emission layer of the light-emitting device may include a dopant and a host, and the dopant may include the heterocyclic compound. For example, the heterocyclic compound may serve as a dopant. For example, the emission layer may emit blue light. The blue light may have a maximum emission wavelength in a range of, for example, about 430 nm to about 480 nm.
[0118] In embodiments, the electron transport region of the light-emitting device may include a hole blocking layer, and the hole blocking layer may include a phosphine oxide-containing compound, a silicon-containing compound, or any combination thereof. For example, the hole blocking layer may contact (e.g., directly contact) the emission layer.
[0119] In an embodiment, the light-emitting device may further include: a second compound including at least one π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group; or a third compound including a group represented by Formula 3; or a fourth compound that emits delayed fluorescence, or any combination thereof, and

[0120] the heterocyclic compound, the second compound, the third compound, and the fourth compound may be different from each other;
##STR00004##
[0121] In Formula 3, [0122] ring CY.sub.71 and ring CY.sub.72 may each independently be a π electron-rich C.sub.3-C.sub.60 cyclic group or a pyridine group, [0123] X.sub.71 may be: a single bond; or a linking group including O, S, N, B, C, Si, or any combination thereof, and [0124] * indicates a binding site to any atom included in a remaining part of the third compound other than the group represented by Formula 3.
[0125] In an embodiment, the heterocyclic compound may include at least one deuterium.
[0126] In embodiments, the second compound, the third compound, and the fourth compound may each independently include at least one deuterium.
[0127] In embodiments, the second compound may include at least one silicon.
[0128] In embodiments, the third compound may include at least one silicon.
[0129] In embodiments, the light-emitting device may further include the second compound and the third compound, in addition to the heterocyclic compound, and at least one of the second compound and the third compound may each independently include at least one deuterium, at least one silicon, or any combination thereof.
[0130] In an embodiment, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the second compound, in addition to the heterocyclic compound. At least one of the heterocyclic compound and the second compound may each independently include at least one deuterium. For example, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the third compound, the fourth compound, or any combination thereof, in addition to the heterocyclic compound and the second compound.
[0131] In embodiments, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the third compound, in addition to the heterocyclic compound. At least one of the heterocyclic compound and the third compound may each independently include at least one deuterium. For example, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the second compound, the fourth compound, or any combination thereof, in addition to the heterocyclic compound and the third compound.
[0132] In embodiments, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the fourth compound, in addition to the heterocyclic compound. At least one of the heterocyclic compound and the fourth compound may each independently include at least one deuterium. The fourth compound may improve the color purity, luminescence efficiency, and lifespan characteristics of the light-emitting device. For example, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the second compound, the third compound, or any combination thereof, in addition to the heterocyclic compound and the fourth compound.
[0133] In embodiments, the light-emitting device (e.g., the emission layer in the light-emitting device) may further include the second compound and the third compound, in addition to the heterocyclic compound. The second compound and the third compound may form an exciplex. At least one of the heterocyclic compound, the second compound, and the third compound may each independently include at least one deuterium.
[0134] In embodiments, the emission layer of the light-emitting device may include: the heterocyclic compound; and the second compound, the third compound, the fourth compound, or any combination thereof; and the emission layer may emit blue light.
[0135] In embodiments, the blue light may have a maximum emission wavelength in a range of about 430 nm to about 480 nm. For example, the blue light may have a maximum emission wavelength in a range of about 430 nm to about 475 nm. For example, the blue light may have a maximum emission wavelength in a range of about 440 nm to about 475 nm. For example, the blue light may have a maximum emission wavelength in a range of about 450 nm to about 475 nm. For

example, the blue light may have a maximum emission wavelength in a range of about 430 nm to about 470 nm. For example, the blue light may have a maximum emission wavelength in a range of about 440 nm to about 470 nm. For example, the blue light may have a maximum emission wavelength in a range of about 450 nm to about 470 nm. For example, the blue light may have a maximum emission wavelength in a range of about 430 nm to about 465 nm. For example, the blue light may have a maximum emission wavelength in a range of about 440 nm to about 465 nm. For example, the blue light may have a maximum emission wavelength in a range of about 450 nm to about 465 nm. For example, the blue light may have a maximum emission wavelength in a range of about 430 nm to about 460 nm. For example, the blue light may have a maximum emission wavelength in a range of about 440 nm to about 460 nm. For example, the blue light may have a maximum emission wavelength in a range of about 450 nm to about 460 nm.
[0136] In an embodiment, the second compound may include a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or any combination thereof.
[0137] In embodiments, the third compound may not include CBP or mCBP:
##STR00005##
[0138] In an embodiment, the fourth compound may be a compound in which a difference between a triplet energy level (eV) and a singlet energy level (eV) may be in a range of about 0 eV to about 0.5 eV. For example, the fourth compound may be a compound in which a difference between a triplet energy level (eV) and a singlet energy level (eV) may be in a range of about 0 eV to about 0.3 eV.
[0139] In embodiments, the fourth compound may be a compound including at least one cyclic group including both boron (B) and nitrogen (N) as ring-forming atoms.
[0140] In embodiments, the fourth compound may be a C.sub.8-C.sub.60 polycyclic group-containing compound including two or more cyclic groups that are condensed with each other while sharing B.
[0141] In embodiments, the fourth compound may include a condensed ring in which at least one third ring is condensed with at least one fourth ring,
[0142] wherein the third ring may be a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclopentene group, a cyclohexene group, a cycloheptene group, a cyclooctene group, an adamantane group, a norbornene group, a norbornane group, a bicyclo[1.1.1] pentane group, a bicyclo[2.1.1] hexane group, a bicyclo[2.2.2] octane group, a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, or a triazine group, and
[0143] the fourth ring may be a 1,2-azaborinine group, a 1,3-azaborinine group, a 1,4-azaborinine group, a 1,2-dihydro-1,2-azaborinine group, a 1,4-oxaborinine group, a 1,4-thiaborinine group, or a 1,4-dihydroborinine group.
[0144] In an embodiment, the third compound may not include a compound represented by Formula 3-1 as described herein.
[0145] In an embodiment, the second compound may include a compound represented by Formula 2:
##STR00006##
[0146] In Formula 2, [0147] L.sub.51 to L.sub.53 may each independently be a single bond, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0148] b51 to b53 may each independently be an integer from 1 to 5, [0149] X.sub.54 may be N or C(R.sub.54), X.sub.55 may be N or C(R.sub.55), X.sub.56 may be N or C(R.sub.56), and at least one of X.sub.54 to X.sub.56 may each be N, and [0150] R.sub.51 to R.sub.56 and R.sub.10a may each be the same as described herein.
[0151] In embodiments, the third compound may include a compound represented by Formula 3-1, a compound represented by Formula 3-2, a compound represented by Formula 3-3, a compound represented by Formula 3-4, a compound represented by Formula 3-5, or any combination thereof:

##STR00007## ##STR00008##
[0152] In Formulae 3-1 to 3-5, [0153] ring CY.sub.71 to ring CY.sub.74 may each independently be a π electron-rich C.sub.3-C.sub.60 cyclic group or a pyridine group, [0154] X.sub.82 may be a single bond, O, S, N—[(L.sub.82).sub.b82-R.sub.82], C(R.sub.82a)(R.sub.82b), or Si(R.sub.82a)(R.sub.82b), [0155] X.sub.83 may be a single bond, O, S, N—[(L.sub.83).sub.b83-R.sub.83], C(R.sub.83a)(R.sub.83b), or Si(R.sub.83a)(R.sub.83b), [0156] X.sub.84 may be O, S, N—[(L.sub.84).sub.b84-R.sub.84], C(R.sub.84a)(R.sub.84b), or Si(R.sub.84a)(R.sub.84b), [0157] X.sub.85 may be C or Si, [0158] L.sub.81 to L.sub.85 may each independently be a single bond, *—C(Q.sub.4)(Q.sub.5)—**, *—Si(Q.sub.4)(Q.sub.5)—**, a π electron-rich C.sub.3-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a, or a pyridine group unsubstituted or substituted with at least one R.sub.10a, wherein Q.sub.4 and Q.sub.5 may each independently be the same as described in connection with Q.sub.1, [0159] b81 to b85 may each independently be an integer from 1 to 5, [0160] R.sub.71 to R.sub.74, R.sub.81 to R.sub.85, R.sub.82a, R.sub.82b, R.sub.83a, R.sub.83b, R.sub.84a, and R.sub.84b are each the same as described herein, [0161] a71 to a74 may each independently be an integer from 0 to 20, and [0162] R.sub.10a is the same as described herein.
[0163] In embodiments, the fourth compound may include a compound represented by Formula 502, a compound represented by Formula 503, or any combination thereof:
##STR00009##
[0164] In Formulae 502 and 503, [0165] ring A.sub.501 to ring A.sub.504 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, [0166] Y.sub.505 may be O, S, N(R.sub.505), B(R.sub.505), C(R.sub.505a)(R.sub.505b), or Si(R.sub.505a)(R.sub.505b), [0167] Y.sub.506 may be O, S, N(R.sub.506), B(R.sub.506), C(R.sub.506a)(R.sub.506b), or Si(R.sub.506a)(R.sub.506b), [0168] Y.sub.507 may be O, S, N(R.sub.507), B(R.sub.507), C(R.sub.507a)(R.sub.507b), or Si(R.sub.507a)(R.sub.507b), [0169] Y.sub.508 may be O, S, N(R.sub.508), B(R.sub.508), C(R.sub.508a)(R.sub.508b), or Si(R.sub.508a)(R.sub.508b), [0170] Y.sub.51 and Y.sub.52 may each independently be B, P(═O), or S(═O), [0171] R.sub.500a, R.sub.500b, R.sub.501 to R.sub.508, R.sub.505a, R.sub.505b, R.sub.506a, R.sub.506b, R.sub.507a, R.sub.507b, R.sub.508a, and R.sub.508b are each the same as described herein, and [0172] a501 to a504 may each independently be an integer from 0 to 20.
[0173] The light-emitting device may have a structure according to a first embodiment or a second embodiment.
First Embodiment
[0174] According to a first embodiment, the heterocyclic compound may be included in the emission layer in the interlayer of the light-emitting device, wherein the emission layer may further include a host. The heterocyclic compound and the host may be different from each other, and the emission layer may emit phosphorescence or fluorescence (e.g., delayed fluorescence) emitted from the heterocyclic compound. For example, according to the first embodiment, the heterocyclic compound may be a dopant or an emitter. For example, the heterocyclic compound may be a fluorescent dopant or a fluorescence emitter.
[0175] Phosphorescence or fluorescence emitted from the heterocyclic compound may be blue light.
[0176] The emission layer may further include an auxiliary dopant. The auxiliary dopant may improve the luminescence efficiency from the heterocyclic compound by effectively transferring energy to the heterocyclic compound, which may be a dopant or an emitter.
[0177] The auxiliary dopant may be different from each of the heterocyclic compound and the host.
[0178] In an embodiment, the auxiliary dopant may be a compound emitting delayed fluorescence.
[0179] In embodiments, the auxiliary dopant may be a compound that includes at least one cyclic group including both boron (B) and nitrogen (N) as ring-forming atoms.
Second Embodiment

[0180] According to a second embodiment, the heterocyclic compound may be included in the emission layer in the interlayer of the light-emitting device, wherein the emission layer may further include a host and a dopant. The heterocyclic compound, the host, and the dopant may be different from each other, and the emission layer may emit phosphorescence or fluorescence (e.g., delayed fluorescence) emitted from the dopant.
[0181] In an embodiment, the heterocyclic compound in the second embodiment may not serve as a dopant, and may serve as an auxiliary dopant or as a sensitizer that transfers energy to a dopant (or emitter).
[0182] In embodiments, the heterocyclic compound in the second embodiment may serve as an emitter and may also serve as an auxiliary dopant (or sensitizer) that transfers energy to a dopant (or emitter).
[0183] For example, phosphorescence or fluorescence emitted from the dopant (or emitter) in the second embodiment may be blue phosphorescence or blue fluorescence (e.g., blue delayed fluorescence).
[0184] The dopant (or emitter) in the second embodiment may be a phosphorescent dopant material (e.g., an organometallic compound represented by Formula 401) or any fluorescent dopant material (e.g., the heterocyclic compound, a compound represented by Formula 501, a compound represented by Formula 502, a compound represented by Formula 503, or any combination thereof).
[0185] The blue light in the first embodiment and the second embodiment may be blue light having a maximum emission wavelength in a range of about 390 nm to about 500 nm. For example, the blue light may have a maximum emission wavelength in a range of about 410 nm to about 490 nm. For example, the blue light may have a maximum emission wavelength in a range of about 430 nm to about 480 nm. For example, the blue light may have a maximum emission wavelength in a range of about 440 nm to about 475 nm. For example, The blue light may have a maximum emission wavelength in a range of about 455 nm to about 470 nm.
[0186] The auxiliary dopant or the sensitizer in the first embodiment may include, for example, the fourth compound represented by Formula 502 or Formula 503.
[0187] In an embodiment, the host in the first embodiment and the second embodiment may be any host material (e.g., a compound represented by Formula 301, a compound represented by 301-1, a compound represented by Formula 301-2, or any combination thereof).
[0188] In embodiments, the host in the first embodiment and the second embodiment may be the second compound, the third compound, or any combination thereof.
[0189] In embodiments, the light-emitting device may further include a capping layer outside the first electrode and/or a capping layer outside the second electrode.
[0190] In embodiments, the light-emitting device may further include at least one of a first capping layer outside the first electrode and a second capping layer outside the second electrode, and the heterocyclic compound may be included in at least one of the first capping layer and the second capping layer. The first capping layer and/or the second capping layer may be the same as described herein.
[0191] In an embodiment, the light-emitting device may include: [0192] a first capping layer outside the first electrode and including the heterocyclic compound; [0193] a second capping layer outside the second electrode and including the heterocyclic compound; or [0194] the first capping layer and the second capping layer.
[0195] The expression "(an interlayer and/or a capping layer) includes a heterocyclic compound represented by Formula 1" as used herein may include an embodiment in which "(an interlayer and/or a capping layer) includes identical heterocyclic compounds represented by Formula 1" and an embodiment in which "(an interlayer and/or a capping layer) includes two or more different heterocyclic compounds each independently represented by Formula 1."
[0196] In an embodiment, the interlayer and/or the capping layer may include, as the heterocyclic

compound, Compound 1 only. For example, Compound 1 may be present in the emission layer of the light-emitting device. In embodiments, the interlayer may include, as the heterocyclic compound, Compounds 1 and 2. For example, Compounds 1 and 2 may be present in a same layer (e.g., both Compounds 1 and 2 may be present in the emission layer), or may be present in different layers (e.g., Compound 1 may be present in the emission layer, and Compound 2 may be present in the electron transport region).
[0197] The term "interlayer" as used herein refers to a single layer and/or all layers between the first electrode and the second electrode of the light-emitting device.
[0198] According to an embodiment, an electronic apparatus may include the light-emitting device. The electronic apparatus may further include a thin-film transistor. In an embodiment, the electronic apparatus may further include a thin-film transistor including a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically connected to at least one of the source electrode and the drain electrode. In an embodiment, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof. The electronic apparatus may be the same as described herein.
[0199] According to an embodiment, an electronic equipment may include the light-emitting device.
[0200] In an embodiment, the electronic equipment may be a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.
[0201] Embodiments provide a heterocyclic compound, which may be the same as described herein.
[0202] Synthesis methods of the heterocyclic compound may be recognizable by one of ordinary skill in the art by referring to the Synthesis Examples and/or the Examples provided below.
[0203] [Description of heterocyclic compound]
[0204] According to an embodiment, a heterocyclic compound may include: an electron donor moiety including nitrogen; an electron acceptor moiety including boron; and a spiro core atom, wherein [0205] the spiro core atom may be bonded to the electron donor moiety via a first bond and a second bond, and the spiro core atom may be bonded to the electron acceptor moiety via a third bond and a fourth bond, [0206] the first bond, the second bond, the third bond, and the fourth bond may each be a single bond, and at least one of Conditions 1 to 3 may be satisfied:
[Condition 1]
[0207] the electron donor moiety includes a group represented by Formula 1A;
[Condition 2]
[0208] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the first ring and the second ring are linked to each other via a first linking group;
[Condition 3]
[0209] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the second ring and the third ring are linked to each other via a second linking group;
##STR00010##
[0210] In Formula 1A, [0211] R.sub.61 to R.sub.63 may each independently be hydrogen,

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), [0212] a61 and a63 may each independently be an integer from 1 to 5, [0213] a62 may be an integer from 1 to 3, [0214] * indicates a binding site to a neighboring atom, [0215] R.sub.10a may be: [0216] hydrogen, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; [0217] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.5-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or any combination thereof; [0218] a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or any combination thereof; or [0219] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32), and [0220] Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
[0221] In Formula 1A, a61 to a63 respectively indicate the number of R.sub.61 to the number of R.sub.63. When a61 is 2 or more, two or more of R.sub.61 may be identical to or different from each other, when a62 is 2 or more, two or more of R.sub.62 may be identical to or different from each other, and when a63 is 2 or more, two or more of R.sub.63 may be identical to or different from each other.
[0222] In an embodiment, the heterocyclic compound may satisfy Conditions 2 and 3.
[0223] In an embodiment, the heterocyclic compound may satisfy all of Conditions 1 to 3.
[0224] In an embodiment, the spiro core atom may be C, Si, or Ge.
[0225] In an embodiment, the spiro core atom may be Si or Ge.
[0226] In Conditions 2 and 3, [0227] the first linking group may be *—C(Z.sub.1)(Z.sub.2)—**, *—Si(Z.sub.1)(Z.sub.2)—**, *—N(Z.sub.1)—**, *—O—*′, or *—S—*′, [0228] the second linking group may be *—C(Z.sub.3)(Z.sub.4)—**, *—Si(Z.sub.3)(Z.sub.4)—**, *—N(Z.sub.3)—

, *—O— or *—S—**, [0229] Z.sub.1 to Z.sub.4 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), [0230] Z.sub.1 and Z.sub.2 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and [0231] Z.sub.3 and Z.sub.4 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a. [0232] R.sub.10a and Q.sub.1 to Q.sub.3 may each be the same as described herein.

[0233] In an embodiment, the heterocyclic compound may be represented by Formula 1: ##STR00011## [0234] In Formula 1, [0235] T.sub.1 may be C, Si, or Ge, [0236] ring CY.sub.1 to ring CY.sub.5 may each independently be a C.sub.5-C.sub.80 carbocyclic group or a C.sub.1-C.sub.30 heterocyclic group, [0237] T.sub.21 may be *—C(Z.sub.1)(Z.sub.2)—*, *—Si(Z.sub.1)(Z.sub.2)—*, *—N(Z.sub.1)—*, *—O—*, or *—S—*, T.sub.22 may be *—C(Z.sub.3)(Z.sub.4)—**, *—Si(Z.sub.3)(Z.sub.4)—**, *—N(Z.sub.3)—**, *—O—**, or *—S—**, [0238] R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.5-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), [0239] a1 to a5 may each independently be an integer from 1 to 20, [0240] Ar.sub.1 may be a group represented by Formula 1A, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0241] Z.sub.1 and Z.sub.2 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0242] Z.sub.3 and Z.sub.4 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0243] two or more neighboring groups among R.sub.1 in the number of a1, R.sub.2 in the number of a2, R.sub.3 in the number of a3, R.sub.4 in the number of a4, R.sub.5 in the number of a5, Z.sub.1 to Z.sub.4, and Ar.sub.1 may optionally be bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and [0244] R.sub.10a and Q.sub.1 to Q.sub.3 may each be the same as described herein.

[0245] In an embodiment, T.sub.1 may be Si or Ge.

[0246] In an embodiment, ring CY.sub.1 to ring CY.sub.5 may each independently be: [0247] a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an azafluorene group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a dibenzooxasiline group, a dibenzothiasiline group, a dibenzodihydroazasiline group, a dibenzodihydrodihydrodisiline group, a dibenzodihydrosiline group, a dibenzodioxine group, a dibenzooxathiine group, a dibenzooxazine group, a dibenzopyran group, a dibenzodithiine group, a dibenzothiazine group, a dibenzothiopyran group, a dibenzocyclohexadiene group, a dibenzodihydropyridine group, or a dibenzodihydropyrazine group.

[0248] In an embodiment, ring CY.sub.1 to ring CY.sub.5 may each independently be a benzene group, a pyridine group, a pyrimidine group, a naphthalene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, a fluorene group, or a dibenzosilole group.

[0249] In an embodiment, ring CY.sub.1 to ring CY.sub.5 may each be a benzene group.

[0250] In an embodiment, T.sub.21 may be *—C(Z.sub.1)(Z.sub.2)—** or *—Si(Z.sub.1)(Z.sub.2)—**, and

[0251] T.sub.22 may be *—C(Z.sub.3)(Z.sub.4)—** or *—Si(Z.sub.3)(Z.sub.4)—**.

[0252] In an embodiment, the heterocyclic compound represented by Formula 1 may satisfy at least one of Conditions 4 and 5:

[Condition 4]

[0253] Z.sub.1 and Z.sub.2 are bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a;

[Condition 5]

[0254] Z.sub.3 and Z.sub.4 are bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

[0255] In an embodiment, the heterocyclic compound represented by Formula 1 may satisfy both Conditions 4 and 5.

[0256] In an embodiment, R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 may each independently be: [0257] hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; [0258] a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof; [0259] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, or an azadibenzosilolyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof; or [0260] —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), and [0261] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

[0262] In an embodiment, R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 may each independently be: [0263] hydrogen, deuterium, —F, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; [0264] a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof; [0265] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with deuterium, —F, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), or any combination thereof; or —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —N(Q.sub.1)(Q.sub.2), and [0266] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

[0267] In an embodiment, R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 may each independently be: [0268] hydrogen, deuterium, or a C.sub.1-C.sub.20 alkyl group; [0269] a C.sub.1-C.sub.20 alkyl group substituted with deuterium, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, or any combination thereof; [0270] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, or a naphthyl group, each unsubstituted or substituted with deuterium, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, a C.sub.1-C.sub.20 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), or any combination thereof; or —C(Q.sub.1)(Q.sub.2)(Q.sub.3) or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), and [0271] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; or a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with deuterium, a C.sub.1-C.sub.60 alkyl group, a phenyl group, a biphenyl group, or any combination thereof.

[0272] In Formula 1, a1 to a5 respectively indicate the number of R.sub.1 to the number of R.sub.5. When a1 is 2 or more, two or more of R.sub.1 may be identical to or different from each other, when a2 is 2 or more, two or more of R.sub.2 may be identical to or different from each other, when a3 is 2 or more, two or more of R.sub.3 may be identical to or different from each other, when a4 is 2 or more, two or more of R.sub.4 may be identical to or different from each other, and

when a5 is 2 or more, two or more of R.sub.5 may be identical to or different from each other.
[0273] In an embodiment, Ar.sub.1 may be: [0274] a group represented by Formula 1A; or [0275] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, or an azadibenzosilolyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof, and [0276] Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
[0277] In an embodiment, Ar.sub.1 may be: [0278] a group represented by Formula 1A; or [0279] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, or a naphthyl group, each unsubstituted or substituted with deuterium, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, a C.sub.1-C.sub.20 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, —

Si(Q.sub.31)(Q.sub.32)(Q.sub.33), or any combination thereof, and [0280] Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; or a C.sub.5-C.sub.60 carbocyclic group unsubstituted or substituted with deuterium, a C.sub.1-C.sub.60 alkyl group, a phenyl group, a biphenyl group, or any combination thereof.

[0281] In an embodiment, Ar.sub.1 may be a group represented by Formula 1A.

[0282] In an embodiment, T.sub.21 and T.sub.22 may each independently be a group represented by Formula 1B:

##STR00012## [0283] In Formula 1B, [0284] R.sub.10a may be the same as described herein, b1 and b2 may each independently be an integer from 1 to 4, [0285] T.sub.3 may be a single bond, *—O—**, *—S—**, *—N(R′)—**, *—C(R′)(R″)—**, or *—Si(R′)(R″)—**, [0286] R′ and R″ may each independently be the same as described in connection with Q.sub.1, and [0287] * and *′ each independently indicate a binding site to a neighboring atom.

[0288] In Formula 1B, b1 and b2 each independently indicate the number of R.sub.10a. When b1 is 2 or more, two or more of R.sub.10a may be identical to or different from each other with respect to (R.sub.10a).sub.b1, and when b2 is 2 or more, two or more of R.sub.10a may be identical to or different from each other with respect to (R.sub.10a).sub.b2.

[0289] In an embodiment, the heterocyclic compound represented by Formula 1 may be represented by Formula 1-1:

##STR00013##

[0290] In Formula 1-1, [0291] T.sub.1, T.sub.21, T.sub.22, and Ar.sub.1 may each be the same as described herein, [0292] X.sub.11 may be C(R.sub.11) or N, X.sub.12 may be C(R.sub.12) or N, X.sub.13 may be C(R.sub.13) or N, and X.sub.14 may be C(R.sub.14) or N, [0293] X.sub.21 may be C(R.sub.21) or N, X.sub.22 may be C(R.sub.22) or N, and X.sub.23 may be C(R.sub.23) or N, [0294] X.sub.31 may be C(R.sub.31) or N, X.sub.32 may be C(R.sub.32) or N, and X.sub.33 may be C(R.sub.33) or N, [0295] X.sub.41 may be C(R.sub.41) or N, X.sub.42 may be C(R.sub.42) or N, and X.sub.43 may be C(R.sub.43) or N, [0296] X.sub.51 may be C(R.sub.51) or N, X.sub.52 may be C(R.sub.52) or N, X.sub.53 may be C(R.sub.53) or N, and X.sub.54 may be C(R.sub.54) or N, [0297] R.sub.11 to R.sub.14 may each independently be the same as described in connection with R.sub.1, [0298] R.sub.21 to R.sub.23 may each independently be the same as described in connection with R.sub.2, [0299] R.sub.31 to R.sub.33 may each independently be the same as described in connection with R.sub.3, [0300] R.sub.41 to R.sub.43 may each independently be the same as described in connection with R.sub.4, [0301] R.sub.51 to R.sub.54 may each independently be the same as described in connection with R.sub.5, and [0302] two or more neighboring groups among R.sub.11 to R.sub.14, R.sub.21 to R.sub.23, R.sub.31 to R.sub.33, R.sub.41 to R.sub.43, R.sub.51 to R.sub.54, Z.sub.1 to Z.sub.4, and Ar.sub.1 may optionally be bonded to each other to form a

[0303] C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

[0304] In Formula 1, unless defined otherwise, R.sub.10a may be: [0305] hydrogen, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; [0306] a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or any combination thereof; [0307] a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-

C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or any combination thereof; or [0308] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32).
[0309] In Formula 1, unless defined otherwise, Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
[0310] The heterocyclic compound may include an electron donor moiety including nitrogen, an electron acceptor moiety including boron, and a spiro core atom, wherein the spiro core atom may be bonded to the electron donor moiety via a first bond and a second bond, and the spiro core atom may be bonded to the electron acceptor moiety via a third bond and a fourth bond, the first bond, the second bond, the third bond, and the fourth bond may each be a single bond, and the heterocyclic compound may satisfy at least one of Conditions 1 to 3 as described herein.
[0311] In thermally activated delayed fluorescence (TADF), slow reverse intersystem crossing (RISC) as a speed-determining process may slow down the entire luminescence system. To achieve high-speed RISC, spatial separation of a highest occupied molecular orbital (HOMO) and a lowest unoccupied molecular orbital (LUMO) may be implemented, and spin-orbit coupling (SOC) may be strengthened. SOC may be strengthened through the heavy atom effect.
[0312] According to embodiments, spatial separation of the electron donor moiety and the electron acceptor moiety may be achieved within the molecule, and the electron donor moiety and the electron acceptor moiety may be orthogonal to each other via the spiro core atom. Accordingly, a highest occupied molecular orbital (HOMO) and a lowest unoccupied molecular orbital (LUMO) may be separated to generate TADF, and internal quantum efficiency may be improved due to high-speed RISC. In an embodiment, C, Si, Ge, and the like may be used as examples of the spiro core atom. In an embodiment, SOC may be strengthened due to the heavy atom effect, thereby accelerating RISC. In an embodiment, due to excellent high-temperature durability, thermal decomposition may not readily occur even under high-temperature conditions, thereby contributing to long lifespan of a device.
[0313] In an embodiment, when Conditions 1 to 3 as described herein are satisfied, the heterocyclic compound may include at least one bulky insulating site in the molecule, and unfavorable transfers, such as Dexter energy transfer (DET), may be suppressed.
[0314] Accordingly, by using the heterocyclic compound, an electronic device (e.g., an organic light-emitting device) having reduced driving voltage, improved color purity and efficiency, and increased lifespan may be implemented.
[Descriptions of Formulae 2, 3, 3-1 to 3-5, 502, and 503]
##STR00014##
[0315] In Formula 2, L.sub.51 to L.sub.53 may each independently be a single bond, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.
[0316] In Formula 2, b51 to b53 respectively indicate the number of L.sub.51 to L.sub.53, and b51 to b53 may each independently be an integer from 1 to 5. When b51 is 2 or more, two or more of L.sub.51 may be identical to or different from each other, when b52 is 2 or more, two or more of L.sub.52 may be identical to or different from each other, and when b53 is 2 or more, two or more

of L.sub.53 may be identical to or different from each other. For example, b51 to b53 may each independently be **1** or **2**.
[0317] In an embodiment, in Formula 2, L.sub.51 to L.sub.53 may each independently be: [0318] a single bond; or [0319] a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an azafluorene group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a dibenzooxacilline group, a dibenzothiacilline group, a dibenzodihydroazacilline group, a dibenzodihydrodicilline group, a dibenzodihydrocilline group, a dibenzodioxane group, a dibenzooxathiene group, a dibenzooxazine group, a dibenzopyran group, a dibenzodithiine group, a dibenzothiazine group, a dibenzothiopyran group, a dibenzocyclohexadiene group, a dibenzodihydropyridine group, or a dibenzodihydropyrazine group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, a triazinyl group, a fluorenyl group, a dimethylfluorenyl group, a diphenylfluorenyl group, a carbazolyl group, a phenylcarbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a dimethyldibenzosilolyl group, a diphenyldibenzosilolyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof, and [0320] Q.sub.31 to Q.sub.33 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, or a triazinyl group.
[0321] In an embodiment, in Formula 2, a bond between L.sub.51 and R.sub.51, a bond between L.sub.52 and R.sub.52, a bond between L.sub.53 and R.sub.53, a bond between two or more of L.sub.51, a bond between two or more of L.sub.52, a bond between two or more of L.sub.53, a bond between L.sub.51 and carbon between X.sub.54 and X.sub.55, a bond between L.sub.52 and carbon between X.sub.54 and X.sub.56, and a bond between L.sub.53 and carbon between X.sub.55 and X.sub.56 may each be a “carbon-carbon single bond.”
[0322] In Formula 2, X.sub.54 may be N or C(R.sub.54), X.sub.55 may be N or C(R.sub.55), and X.sub.56 may be N or C(R.sub.56), wherein at least one of X.sub.54 to X.sub.56 may each be N. R.sub.54 to R.sub.56 may each be the same as described herein. In an embodiment, two or three of X.sub.54 to X.sub.56 may each be N.
[0323] In Formula 2, R.sub.51 to R.sub.56 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group

unsubstituted or substituted with at least one R.sub.10a, a C.sub.7-C.sub.60 arylalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 heteroarylalkyl group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2). Q.sub.1 to Q.sub.3 may each be the same as described herein.

[0324] In an embodiment, in Formula 2, R.sub.51 to R.sub.56 may each independently be: [0325] hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; [0326] a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof; [0327] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, an azadibenzosilolyl group, or a group represented by Formula 91, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)

(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof; or [0328] —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), and [0329] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: [0330] CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; or [0331] an n-propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, or a triazinyl group, each unsubstituted or substituted with deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, a triazinyl group, or any combination thereof;
##STR00015##
[0332] In Formula 91, [0333] ring CY.sub.91 and ring CY.sub.92 may each independently be a C.sub.5-C.sub.30 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.30 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0334] X.sub.91 may be a single bond, O, S, N(R.sub.91), B(R.sub.91), C(R.sub.91a)(R.sub.91b), or Si(R.sub.91a)(R.sub.91b), [0335] R.sub.91, R.sub.91a, and R.sub.91b may respectively be the same as described in connection with R.sub.82, R.sub.82a, and R.sub.82b, as described herein, [0336] R.sub.10a may be the same as described herein, and [0337] * indicates a binding site to a neighboring atom.
[0338] In an embodiment, in Formula 91, [0339] ring CY.sub.91 and ring CY.sub.92 may each independently be a benzene group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, or a triazine group, each unsubstituted or substituted with at least one R.sub.10a, and [0340] R.sub.91, R.sub.91a, and R.sub.91b may each independently be: [0341] hydrogen or a C.sub.1-C.sub.10 alkyl group; or [0342] a phenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, or a triazinyl group, each unsubstituted or substituted with deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, a triazinyl group, or any combination thereof.
[0343] In an embodiment, in Formula 2, a group represented by *—(L.sub.51).sub.b51-R.sub.51 and a group represented by *—(L.sub.52).sub.b52-R.sub.52 may each not be a phenyl group.
[0344] In an embodiment, in Formula 2, a group represented by *—(L.sub.51).sub.b51-R.sub.51 and a group represented by *—(L.sub.52).sub.b52-R.sub.52 may be identical to each other.
[0345] In embodiments, in Formula 2, a group represented by *—(L.sub.51).sub.b51-R.sub.51 and a group represented by *—(L.sub.52).sub.b52-R.sub.52 may be different from each other.
[0346] In embodiments, in Formula 2, R.sub.51 and R.sub.52 may each independently be 1, 2, or 3; and L.sub.51 and L.sub.52 may each independently be a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, or a triazine group, each unsubstituted or substituted with at least one R.sub.10a.
[0347] In an embodiment, in Formula 2, R.sub.51 and R.sub.52 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), and
[0348] Q.sub.1 to Q.sub.3 may each independently be a C.sub.8-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a

cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

[0349] In an embodiment, in Formula 2, [0350] a group represented by *—(L.sub.51).sub.b51-R.sub.51 may be a group represented by one of Formulae CY51-1 to CY51-26, and/or [0351] a group represented by *—(L.sub.52).sub.b52-R.sub.52 may be a group represented by one of Formulae CY52-1 to CY52-26, and/or [0352] a group represented by *—(L.sub.53).sub.b53-R.sub.53 may be a group represented by one of Formulae CY53-1 to CY53-27, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3):

##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027##

[0353] In Formulae CY51-1 to CY51-26, CY52-1 to CY52-26, and CY53-1 to CY53-27, [0354] Y.sub.63 may be a single bond, O, S, N(R.sub.63), B(R.sub.63), C(R.sub.63a)(R.sub.63a), or Si(R.sub.63a)(R.sub.63a), [0355] Y.sub.64 may be a single bond, O, S, N(R.sub.64), B(R.sub.64), C(R.sub.64a)(R.sub.64b), or Si(R.sub.64a)(R.sub.64b), [0356] Y.sub.67 may be a single bond, O, S, N(R.sub.67), B(R.sub.67), C(R.sub.67a)(R.sub.67b), or Si(R.sub.67a)(R.sub.67b), [0357] Y.sub.68 may be a single bond, O, S, N(R.sub.68), B(R.sub.68), C(R.sub.68a)(R.sub.68b), or Si(R.sub.68a)(R.sub.68b), [0358] Y.sub.63 and Y.sub.64 in Formulae CY51-16 and CY51-17 may each not be a single bond at the same time, [0359] Y.sub.67 and Y.sub.68 in Formulae CY52-16 and CY52-17 may each not be a single bond at the same time, [0360] R.sub.51a to R.sub.51e, R.sub.61 to R.sub.64, R.sub.63a, R.sub.63b, R.sub.64a, and R.sub.64b may each independently be the same as described in connection with R.sub.51, except that R.sub.51a to R.sub.51e may each not be hydrogen, [0361] R.sub.52a to R.sub.52e, R.sub.65 to R.sub.68, R.sub.67a, R.sub.67b, R.sub.68a, and R.sub.68b may each independently be the same as described in connection with R.sub.52, except that R.sub.52a to R.sub.52e may each not be hydrogen, [0362] R.sub.53a to R.sub.53e, R.sub.69a, and R.sub.69b may each independently be the same as described in connection with R.sub.53, except that R.sub.53a to R.sub.53e may each not be hydrogen, and [0363] * indicates a binding site to a neighboring atom.

[0364] In an embodiment, [0365] in Formulae CY51-1 to CY51-26 and CY52-1 to 52-26, R.sub.51a to R.sub.51e and R.sub.52a to R.sub.52e may each independently be: [0366] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, an azadibenzosilolyl group, or a group represented by Formula 91, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a

biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, or any combination thereof; or [0367] —C(Q.sub.1)(Q.sub.2)(Q.sub.3) or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), [0368] Q.sub.1 to Q.sub.3 may each independently be a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, or a triazinyl group, each unsubstituted or substituted with deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, a triazinyl group, or any combination thereof, [0369] in Formulae CY51-16 and CY51-17, Y.sub.63 may be O or S, and Y.sub.64 may be Si(R.sub.64a)(R.sub.64b); or Y.sub.63 may be Si(R.sub.63a)(R.sub.63a), and Y.sub.64 may be O or S, and [0370] in Formulae CY52-16 and CY52-17, Y.sub.67 may be O or S, and Y.sub.68 may be Si(R.sub.68a)(R.sub.68b); or Y.sub.67 may be Si(R.sub.67a)(R.sub.67b), and Y.sub.68 may be O or S.
##STR00028##
[0371] In Formula 3, ring CY.sub.71 and ring CY.sub.72 may each independently be a π electron-rich C.sub.3-C.sub.60 cyclic group or a pyridine group.
[0372] In Formula 3, X.sub.71 may be: a single bond; or a linking group including O, S, N, B, C, Si, or any combination thereof.
[0373] In Formula 3, * indicates a binding site to any atom included in a remaining part of the third compound other than the group represented by Formula 3.
##STR00029## ##STR00030##
[0374] In Formulae 3-1 to 3-5, ring CY.sub.71 to ring CY.sub.74 may each independently be a IT electron-rich C.sub.3-C.sub.60 cyclic group or a pyridine group.
[0375] In Formulae 3-1 to 3-5, X.sub.82 may be a single bond, O, S, N—[(L.sub.82) 682-R.sub.82], C(R.sub.82a)(R.sub.82b), or Si(R.sub.82a)(R.sub.82b).
[0376] In Formulae 3-1 to 3-5, X.sub.83 may be a single bond, O, S, N—[(L.sub.83).sub.b83-R.sub.83], C(R.sub.83a)(R.sub.83b), or Si(R.sub.83a)(R.sub.83b).
[0377] In Formulae 3-1 to 3-5, X.sub.84 may be O, S, N—[(L.sub.84)R.sub.84-R.sub.84], C(R.sub.84a)(R.sub.84b), or Si(R.sub.84a)(R.sub.84b).
[0378] In Formulae 3-1 to 3-5, X.sub.85 may be C or Si.
[0379] In Formulae 3-1 to 3-5, L.sub.81 to L.sub.85 may each independently be a single bond, *—C(Q.sub.4)(Q.sub.5)—**, *—Si(Q.sub.4)(Q.sub.5)—**, a π electron-rich C.sub.3-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a, or a pyridine group unsubstituted or substituted with at least one R.sub.10a.
[0380] In Formulae 3-1 to 3-5, Q.sub.4 and Q.sub.5 may each independently be the same as described in connection with Q.sub.1.
[0381] In Formulae 3-1 to 3-5, b81 to b85 may each independently be an integer from 1 to 5.
[0382] In Formulae 3-1 to 3-5, R.sub.71 to R.sub.74, R.sub.81 to R.sub.85, R.sub.82a, R.sub.82b, R.sub.83a, R.sub.83b, R.sub.84a, and R.sub.84b may each be the same as described herein.
[0383] In Formulae 3-1 to 3-5, a71 to a74 respectively indicate the number of R.sub.71 to the number of R.sub.74, and a71 to a74 may each independently be an integer from 0 to 20. When a71

is 2 or more, two or more of R.sub.71 may be identical to or different from each other, when a72 is 2 or more, two or more of R.sub.72 may be identical to or different from each other, when a73 is 2 or more, two or more of R.sub.73 may be identical to or different from each other, and when a74 is 2 or more, two or more of R.sub.74 may be identical to or different from each other. In an embodiment, a71 to a74 may each independently be an integer from 0 to 8.
[0384] In Formulae 3-1 to 3-5, R.sub.10a may be the same as described herein.
[0385] In an embodiment, in Formulae 3-1 to 3-5, L.sub.81 to L.sub.85 may each independently be: [0386] a single bond; or [0387] *—C(Q.sub.4)(Q.sub.5)—** or *—Si(Q.sub.4)(Q.sub.5)—**; or [0388] a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an azafluorene group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, or a benzothiadiazole group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, a triazinyl group, a fluorenyl group, a dimethylfluorenyl group, a diphenylfluorenyl group, a carbazolyl group, a phenylcarbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a dimethyldibenzosilolyl group, a diphenyldibenzosilolyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof, and [0389] Q.sub.4, Q.sub.5, and Q.sub.31 to Q.sub.33 may each independently be hydrogen, deuterium, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, or a triazinyl group.
[0390] In embodiments, in Formulae 3-1 and 3-2, a group represented by
##STR00031##
may be a group represented by one of Formulae CY71-1(1) to CY71-1(8), and/or
[0391] in Formulae 3-1 and 3-3, a group represented by
##STR00032##
may be a group represented by one of Formulae CY71-2(1) to CY71-2(8), and/or
[0392] in Formulae 3-2 and 3-4, a group represented by
##STR00033##
may be a group represented by one of Formulae CY71-3(1) to CY71-3(32), and/or
[0393] in Formulae 3-3 to 3-5, a group represented by
##STR00034##
may be a group represented by one of Formulae CY71-4(1) to CY71-4(32), and/or
[0394] in Formula 3-5, a group represented by
##STR00035##
represented by one of Formulae CY71-5(1) to CY71-5(**8**):
##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041## ##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047##
[0395] In Formulae CY71-1(1) to CY71-1(8), CY71-2(1) to CY71-2(8), CY71-3(1) to CY71-

3(32), CY71-4(1) to CY71-4(32), and CY71-5(1) to CY71-5(8), [0396] X.sub.82 to X.sub.85, L.sub.81, b.sub.81, R.sub.81, and R.sub.85 may each be the same as described herein, [0397] X.sub.86 may be a single bond, O, S, N(R.sub.86), B(R.sub.86), C(R.sub.86a)(R.sub.86b), or Si(R.sub.86a)(R.sub.86b), [0398] X.sub.87 may be a single bond, O, S, N(R.sub.87), B(R.sub.87), C(R.sub.87a)(R.sub.87b), or Si(R.sub.87a)(R.sub.87b), [0399] in Formulae CY71-1(1) to CY71-1(8) and CY71-4(1) to CY71-4(32), X.sub.86 and X.sub.87 may not each be a single bond at the same time, [0400] X.sub.88 may be a single bond, O, S, N(R.sub.88), B(R.sub.88), C(R.sub.88a)(R.sub.88b), or Si(R.sub.88a)(R.sub.88b), [0401] X.sub.89 may be a single bond, O, S, N(R.sub.89), B(R.sub.89), C(R.sub.89a)(R.sub.89b), or Si(R.sub.89a)(R.sub.89b), [0402] in Formulae CY71-2(1) to CY71-2(8), CY71-3(1) to CY71-3(32), and CY71-5(1) to CY71-5(8), X.sub.88 and X.sub.89 may not each be a single bond at the same time, and [0403] R.sub.86 to R.sub.89, R.sub.86a, R.sub.86b, R.sub.87a, R.sub.87b, R.sub.88a, R.sub.88b, R.sub.89a, and R.sub.89b may each independently be the same as described in connection with R.sub.81.
##STR00048##
[0404] In Formulae 502 and 503, ring A.sub.501 to ring A.sub.504 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group.
[0405] In Formulae 502 and 503, Y.sub.505 may be O, S, N(R.sub.505), B(R.sub.505), C(R.sub.505a)(R.sub.505b), or Si(R.sub.505a)(R.sub.505b).
[0406] In Formulae 502 and 503, Y.sub.506 may be O, S, N(R.sub.506), B(R.sub.506), C(R.sub.506a)(R.sub.506b), or Si(R.sub.506a)(R.sub.506b).
[0407] In Formulae 502 and 503, Y.sub.507 may be O, S, N(R.sub.507), B(R.sub.507), C(R.sub.507a)(R.sub.507b), or Si(R.sub.507a)(R.sub.507b).
[0408] In Formulae 502 and 503, Y.sub.508 may be O, S, N(R.sub.508), B(R.sub.508), C(R.sub.508a)(R.sub.508b), or Si(R.sub.508a)(R.sub.508b).
[0409] In Formulae 502 and 503, Y.sub.51 and Y.sub.52 may each independently be B, P(═O), or S(═O).
[0410] In Formulae 502 and 503, R.sub.500a, R.sub.500b, R.sub.501 to R.sub.508, R.sub.505a, R.sub.505b, R.sub.506a, R.sub.506b, R.sub.507a, R.sub.507b, R.sub.508a, and R.sub.508b may each be the same as described herein.
[0411] In Formulae 502 and 503, a501 to a504 respectively indicate the number of R.sub.501 to the number of R.sub.504, and a501 to a504 may each independently be an integer from 0 to 20. When a501 is 2 or more, two or more of R.sub.501 may be identical to or different from each other, when a502 is 2 or more, two or more of R.sub.502 may be identical to or different from each other, when a503 is 2 or more, two or more of R.sub.503 may be identical to or different from each other, and when a504 is 2 or more, two or more of R.sub.504 may be identical to or different from each other. In an embodiment, a501 to a504 may each independently be an integer from 0 to 8.
[0412] In the specification, R.sub.51 to R.sub.56, R.sub.71 to R.sub.74, R.sub.81 to R.sub.85, R.sub.82a, R.sub.82b, R.sub.83a, R.sub.83b, R.sub.84a, R.sub.84b, R.sub.500a, R.sub.500b, R.sub.501 to R.sub.508, R.sub.505a, R.sub.505b, R.sub.506a, R.sub.506b, R.sub.507a, R.sub.507b, R.sub.508a, and R.sub.508b may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.7-C.sub.60 arylalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 heteroarylalkyl group unsubstituted or

substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2). Q.sub.1 to Q.sub.3 may each be the same as described herein.

[0413] In an embodiment, R.sub.51 to R.sub.56, R.sub.71 to R.sub.74, R.sub.81 to R.sub.85, R.sub.82a, R.sub.82b, R.sub.83a, R.sub.83b, R.sub.84a, R.sub.84b, R.sub.500a, R.sub.500b, R.sub.501 to R.sub.508, R.sub.505a, R.sub.505b, R.sub.506a, R.sub.506b, R.sub.507a, R.sub.507b, R.sub.508a, and R.sub.508b in Formulae 2, 3-1 to 3-5, 502, and 503, and R.sub.10a may each independently be: [0414] hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; [0415] a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or any combination thereof; [0416] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, an azadibenzosilolyl group, or a group represented by Formula 91, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33),

—N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or any combination thereof; or [0417] —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), and [0418] Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 may each independently be: [0419] —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CH.sub.2CH.sub.3, —CH.sub.2CD.sub.3, —CH.sub.2CD.sub.2H, —CH.sub.2CDH.sub.2, —CHDCH.sub.3, —CHDCD.sub.2H, —CHDCDH.sub.2, —CHDCD.sub.3, —CD.sub.2CD.sub.3, —CD.sub.2CD.sub.2H, or —CD.sub.2CDH.sub.2; or [0420] an n-propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, an n-pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, or a triazinyl group, each unsubstituted or substituted with deuterium, a C.sub.1-C.sub.10 alkyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a pyridazinyl group, a pyrazinyl group, a triazinyl group, or any combination thereof.
[0421] In embodiments, R.sub.61 to R.sub.63 in Formula 1A, R.sub.1 to R.sub.5 in Formula 1, R.sub.51 to R.sub.56, R.sub.71 to R.sub.74, R.sub.81 to R.sub.85, R.sub.82a, R.sub.82b, R.sub.83a, R.sub.83b, R.sub.84a, R.sub.84b, R.sub.500a, R.sub.500b, R.sub.501 to R.sub.508, R.sub.505a, R.sub.505b, R.sub.506a, R.sub.506b, R.sub.507a, R.sub.507b, R.sub.508a, and R.sub.508b in Formulae 2, 3-1 to 3-5, 502, and 503, and R.sub.10a may each independently be: [0422] hydrogen, deuterium, —F, a cyano group, a nitro group, —CH.sub.3, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, or —CFH.sub.2; [0423] a group represented by one of Formulae 9-1 to 9-19; or [0424] a group represented by one of Formulae 10-1 to 10-246, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), or —P(═O)(Q.sub.1)(Q.sub.2) (wherein Q.sub.1 to Q.sub.3 may each be the same as described herein).
##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072## ##STR00073## ##STR00074## ##STR00075##
##STR00076## ##STR00077## ##STR00078##
[0425] In Formulae 9-1 to 9-19 and 10-1 to 10-246, * indicates a binding site to a neighboring atom, Ph represents a phenyl group, and TMS represents a trimethylsilyl group.
Examples of Compounds
[0426] In an embodiment, the heterocyclic compound may be one of Compounds 1 to 6:
##STR00079## ##STR00080##
[Description of FIG. **1**]
[0427] FIG. **1** is a schematic cross-sectional view of a light-emitting device **10** according to an embodiment. The light-emitting device **10** may include a first electrode **110**, an interlayer **130**, and a second electrode **150**.
[0428] Hereinafter, the structure of the light-emitting device **10** according to an embodiment and a method of manufacturing the light-emitting device **10** will be described with reference to FIG. **1**.
[First Electrode **110**]
[0429] In FIG. **1**, a substrate may be further included under the first electrode **110** or on the second electrode **150**. In an embodiment, the substrate may be a glass substrate or a plastic substrate. In embodiments, the substrate may be a flexible substrate, and may include plastics with excellent heat resistance and durability, such as polyimide, polyethylene terephthalate (PET), polycarbonate, polyethylene naphthalate, polyarylate (PAR), polyetherimide, or any combination thereof.
[0430] The first electrode **110** may be formed by, for example, depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, a

material for forming the first electrode **110** may be a high-work function material that facilitates injection of holes.
[0431] The first electrode **110** may be a reflective electrode, a transflective electrode, or a transmissive electrode. In an embodiment, when the first electrode **110** is a transmissive electrode, a material for forming the first electrode **110** may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO.sub.2), zinc oxide (ZnO), or any combination thereof. In embodiments, when the first electrode **110** is a transflective electrode or a reflective electrode, a material for forming the first electrode **110** may include magnesium (Mg), silver (Ag), aluminum (AI), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combination thereof.
[0432] The first electrode **110** may have a structure consisting of a single layer, or a structure including multiple layers. For example, the first electrode **110** may have a three-layer structure of ITO/Ag/ITO.
[Interlayer **130**]
[0433] The interlayer **130** may be arranged on the first electrode **110**. The interlayer **130** may include an emission layer.
[0434] The interlayer **130** may further include a hole transport region between the first electrode **110** and the emission layer, and an electron transport region between the emission layer and the second electrode **150**.
[0435] The interlayer **130** may further include, in addition to various organic materials, a metal-containing compound such as an organometallic compound, an inorganic material such as a quantum dot, or the like.
[0436] In an embodiment, the interlayer **130** may include two or more emitting units stacked between the first electrode **110** and the second electrode **150**, and at least one charge generation layer between adjacent units among the two or more emitting units. When the interlayer **130** includes the two or more emitting units and the at least one charge generation layer, the light-emitting device **10** may be a tandem light-emitting device.
[Hole Transport Region in Interlayer **130**]
[0437] The hole transport region may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer including different materials, or a structure including multiple layers including different materials.
[0438] The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof.
[0439] In embodiments, the hole transport region may have a multi-layer structure including a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein the layers of each structure may be stacked from the first electrode **110** in its respective stated order, but the structure of the hole transport region is not limited thereto.
[0440] In embodiments, the hole transport region may include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof:
##STR00081##
[0441] In Formulae 201 and 202, [0442] L.sub.201 to L.sub.204 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a,
[0443] L.sub.205 may be *—O—**, *—S—**, *—N(Q.sub.201)—*′, a C.sub.1-C.sub.20 alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60

heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0444] xa1 to xa4 may each independently be an integer from 0 to 5, [0445] xa5 may be an integer from 1 to 10, [0446] R.sub.201 to R.sub.204 and Q.sub.201 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0447] R.sub.201 and R.sub.202 may optionally be linked to each other via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group (e.g., a carbazole group, etc.) unsubstituted or substituted with at least one R.sub.10a (e.g., Compound HT16, etc.), [0448] R.sub.203 and R.sub.204 may optionally be linked to each other via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group unsubstituted or substituted with at least one R.sub.10a, and [0449] na1 may be an integer from 1 to 4.
[0450] In an embodiment, the compound represented by Formula 201 and the compound represented by Formula 202 may each independently include at least one of groups represented by Formulae CY201 to CY217:
##STR00082## ##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087## ##STR00088##
[0451] In Formulae CY201 to CY217, R.sub.10b and R.sub.10c may each independently be the same as described in connection with R.sub.10a, ring CY201 to ring CY204 may each independently be a C.sub.3-C.sub.20 carbocyclic group or a C.sub.1-C.sub.20 heterocyclic group, and at least one hydrogen in Formulae CY201 to CY217 may be unsubstituted or substituted with R.sub.10a.
[0452] In an embodiment, in Formulae CY201 to CY217, ring CY201 to ring CY204 may each independently be a benzene group, a naphthalene group, a phenanthrene group, or an anthracene group.
[0453] In embodiments, the compound represented by Formula 201 and the compound represented by Formula 202 may each independently include at least one of groups represented by Formulae CY201 to CY203.
[0454] In embodiments, the compound represented by Formula 201 may include at least one of groups represented by Formulae CY201 to CY203 and at least one of groups represented by Formulae CY204 to CY217.
[0455] In embodiments, in Formula 201, xa1 may be 1, R.sub.201 may be a group represented by one of Formulae CY201 to CY203, xa2 may be 0, and R.sub.202 may be a group represented by one of Formulae CY204 to CY207.
[0456] In embodiments, the compound represented by Formula 201 and the compound represented by Formula 202 may each not include groups represented by Formulae CY201 to CY203.
[0457] In embodiments, the compound represented by Formula 201 and the compound represented by Formula 202 may each not include groups represented by Formulae CY201 to CY203, and may each independently include at least one of groups represented by Formulae CY204 to CY217.
[0458] In embodiments, the compound represented by Formula 201 and the compound represented by Formula 202 may each not include groups represented by Formulae CY201 to CY217.
[0459] In an embodiment, the hole transport region may include: one of Compounds HT1 to HT46; m-MTDATA; TDATA; 2-TNATA; NPB(NPD); β-NPB; TPD; spiro-TPD; spiro-NPB; methylated NPB; TAPC; HMTPD; 4,4′,4″-tris(N-carbazolyl)triphenylamine (TCTA); polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA); poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate)(PEDOT/PSS); polyaniline/camphor sulfonic acid (PANI/CSA); polyaniline/poly(4-styrenesulfonate) (PANI/PSS); or any combination thereof:
##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094##

##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099##

[0460] A thickness of the hole transport region may be in a range of about 50 Å to about 10,000 Å. For example, the thickness of the hole transport region may be in a range of about 100 Å to about 4,000 Å. When the hole transport region includes a hole injection layer, a hole transport layer, or any combination thereof, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å. For example, the thickness of the hole injection layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the hole transport layer may be in a range of about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within the ranges described above, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

[0461] The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to a wavelength of light emitted by the emission layer, and the electron blocking layer may block the leakage of electrons from the emission layer to the hole transport region. Materials that may be included in the hole transport region may be included in the emission auxiliary layer and the electron blocking layer.

[p-Dopant]

[0462] The hole transport region may further include, in addition to the materials described above, a charge-generation material for the improvement of conductive properties. The charge-generation material may be uniformly or non-uniformly dispersed in the hole transport region (e.g., in the form of a single layer consisting of a charge-generation material).

[0463] The charge-generation material may be, for example, a p-dopant.

[0464] For example, the p-dopant may have a lowest unoccupied molecular orbital (LUMO) energy level less than or equal to about −3.5 eV.

[0465] In an embodiment, the p-dopant may include a quinone derivative, a cyano group-containing compound, a compound including element EL1 and element EL2, or any combination thereof.

[0466] Examples of a quinone derivative may include TCNQ, F4-TCNQ, and the like.

[0467] Examples of a cyano group-containing compound may include HAT-CN, a compound represented by Formula 221, and the like:

##STR00100##

[0468] In Formula 221,

[0469] R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and

[0470] at least one of R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each substituted with: a cyano group; —F; —Cl; —Br; —I; a C.sub.1-C.sub.20 alkyl group substituted with a cyano group, —F, —Cl, —Br, —I, or any combination thereof; or any combination thereof.

[0471] In the compound including element EL1 and element EL2, element EL1 may be a metal, a metalloid, or any combination thereof, and element EL2 may be a non-metal, a metalloid, or any combination thereof.

[0472] Examples of a metal may include: an alkali metal (e.g., lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), etc.); an alkaline earth metal (e.g., beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), etc.); a transition metal (e.g., titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), manganese (Mn), technetium (Tc), rhenium (Re), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), etc.); a post-transition metal (e.g., zinc (Zn), indium (In), tin (Sn), etc.); a lanthanide metal (e.g., lanthanum (La), cerium (Ce), praseodymium

(Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), etc.); and the like.
[0473] Examples of a metalloid may include silicon (Si), antimony (Sb), tellurium (Te), and the like.
[0474] Examples of a non-metal may include oxygen (O), a halogen (e.g., F, Cl, Br, I, etc.), and the like.
[0475] Examples of a compound including element EL1 and element EL2 may include a metal oxide, a metal halide (e.g., a metal fluoride, a metal chloride, a metal bromide, a metal iodide, etc.), a metalloid halide (e.g., a metalloid fluoride, a metalloid chloride, a metalloid bromide, a metalloid iodide, etc.), a metal telluride, or any combination thereof.
[0476] Examples of a metal oxide may include a tungsten oxide (e.g., WO, W.sub.2O.sub.3, WO.sub.2, WO.sub.3, W.sub.2O.sub.5, etc.), a vanadium oxide (e.g., VO, V.sub.2O.sub.3, VO.sub.2, V.sub.2O.sub.5, etc.), a molybdenum oxide (e.g., MoO, Mo.sub.2O.sub.3, MoO.sub.2, MoO.sub.3, Mo.sub.2O.sub.5, etc.), a rhenium oxide (e.g., ReO.sub.3, etc.), and the like.
[0477] Examples of a metal halide may include an alkali metal halide, an alkaline earth metal halide, a transition metal halide, a post-transition metal halide, a lanthanide metal halide, and the like.
[0478] Examples of an alkali metal halide may include LiF, NaF, KF, RbF, CsF, LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, RbBr, CsBr, Lil, Nal, KI, Rbl, Csl, and the like.
[0479] Examples of an alkaline earth metal halide may include BeF.sub.2, MgF.sub.2, CaF.sub.2, SrF.sub.2, BaF.sub.2, BeCl.sub.2, MgCl.sub.2, CaCl.sub.2), SrCl.sub.2, BaCl.sub.2, BeBr.sub.2, MgBr.sub.2, CaBr.sub.2, SrBr.sub.2, BaBr.sub.2, Bel.sub.2, Mgl.sub.2, Cal.sub.2, Srl.sub.2, Bal.sub.2, and the like.
[0480] Examples of a transition metal halide may include a titanium halide (e.g., TiF.sub.4, TiCl.sub.4, TiBr.sub.4, Til.sub.4, etc.), a zirconium halide (e.g., ZrF.sub.4, ZrCl.sub.4, ZrBr.sub.4, Zrl.sub.4, etc.), a hafnium halide (e.g., HfF.sub.4, HfCl.sub.4, HfBr.sub.4, Hfl.sub.4, etc.), a vanadium halide (e.g., VF.sub.3, VCl.sub.3, VBr.sub.3, Vl.sub.3, etc.), a niobium halide (e.g., NbF.sub.3, NbCl.sub.3, NbBr.sub.3, Nbl.sub.3, etc.), a tantalum halide (e.g., TaF.sub.3, TaCl.sub.3, TaBr.sub.3, Tal.sub.3, etc.), a chromium halide (e.g., CrF.sub.3, CrCl.sub.3, CrBrs, Crl.sub.3, etc.), a molybdenum halide (e.g., MoF.sub.3, MoCl.sub.3, MoBr.sub.3, Mol.sub.3, etc.), a tungsten halide (e.g., WF.sub.3, WCl.sub.3, WBr.sub.3, Wl.sub.3, etc.), a manganese halide (e.g., MnF.sub.2, MnCl.sub.2, MnBr.sub.2, Mnl.sub.2, etc.), a technetium halide (e.g., TcF.sub.2, TcCl.sub.2, TcBr.sub.2, Tcl.sub.2, etc.), a rhenium halide (e.g., ReF.sub.2, ReCl.sub.2, ReBr.sub.2, Rel.sub.2, etc.), an iron halide (e.g., FeF.sub.2, FeCl.sub.2, FeBr.sub.2, Fel.sub.2, etc.), a ruthenium halide (e.g., RuF.sub.2, RuCl.sub.2, RuBr.sub.2, Rul.sub.2, etc.), an osmium halide (e.g., OsF.sub.2, OsCl.sub.2, OsBr.sub.2, Osl.sub.2, etc.), a cobalt halide (e.g., CoF.sub.2, CoCl.sub.2, CoBr.sub.2, Col.sub.2, etc.), a rhodium halide (e.g., RhF.sub.2, RhCl.sub.2, RhBr.sub.2, Rhl.sub.2, etc.), an iridium halide (e.g., IrF.sub.2, IrCl.sub.2, IrBr.sub.2, Irl.sub.2, etc.), a nickel halide (e.g., NiF.sub.2, NiCl.sub.2, NiBr.sub.2, Nil.sub.2, etc.), a palladium halide (e.g., PdF.sub.2, PdCl.sub.2, PdBr.sub.2, Pdl.sub.2, etc.), a platinum halide (e.g., PtF.sub.2, PtCl.sub.2, PtBr.sub.2, Ptl.sub.2, etc.), a copper halide (e.g., CuF, CuCl, CuBr, Cul, etc.), a silver halide (e.g., AgF, AgCl, AgBr, Agl, etc.), a gold halide (e.g., AuF, AuCl, AuBr, Aul, etc.), and the like.
[0481] Examples of a post-transition metal halide may include a zinc halide (e.g., ZnF.sub.2, ZnCl.sub.2, ZnBr.sub.2, Znl.sub.2, etc.), an indium halide (e.g., InI.sub.3, etc.), a tin halide (e.g., SmI.sub.2, etc.), and the like.
[0482] Examples of a lanthanide metal halide may include YbF, YbF.sub.2, YbF.sub.3, SmF.sub.3, YbCl, YbCl.sub.2, YbCl.sub.3, SmCl.sub.3, YbBr, YbBr.sub.2, YbBr.sub.3, SmBr.sub.3, Ybl, Ybl.sub.2, Ybl.sub.3, Sml.sub.3, and the like.
[0483] Examples of a metalloid halide may include an antimony halide (e.g., SbCl.sub.5, etc.) and

the like.
[0484] Examples of a metal telluride may include an alkali metal telluride (e.g., Li.sub.2Te, Na.sub.2Te, K.sub.2Te, Rb.sub.2Te, Cs.sub.2Te, etc.), an alkaline earth metal telluride (e.g., BeTe, MgTe, CaTe, SrTe, BaTe, etc.), a transition metal telluride (e.g., TiTe.sub.2, ZrTe.sub.2, HfTe.sub.2, V.sub.2Te.sub.3, Nb.sub.2Tes, Ta.sub.2Tes, Cr.sub.2Tes, Mo.sub.2, Te.sub.3, W.sub.2Te.sub.3, MnTe, TcTe, ReTe, FeTe, RuTe, OsTe, CoTe, RhTe, IrTe, NiTe, PdTe, PtTe, CuzTe, CuTe, Ag.sub.2Te, AgTe, Au.sub.2Te, etc.), a post-transition metal telluride (e.g., ZnTe, etc.), a lanthanide metal telluride (e.g., LaTe, CeTe, PrTe, NdTe, PmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, LuTe, etc.), and the like.
[Emission Layer in Interlayer **130**]
[0485] When the light-emitting device **10** is a full-color light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer, according to a subpixel. In an embodiment, the emission layer may have a stacked structure of two or more of a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers may contact each other or may be separated from each other to emit white light. In embodiments, the emission layer may include two or more of a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials may be mixed with each other in a single layer to emit white light.
[0486] In an embodiment, the emission layer may include a host and a dopant (or emitter). In embodiments, the emission layer may further include an auxiliary dopant that promotes energy transfer to a dopant (or emitter), in addition to the host and the dopant (or emitter). When the emission layer includes the dopant (or emitter) and the auxiliary dopant, the dopant (or emitter) and the auxiliary dopant may be different from each other.
[0487] The heterocyclic compound as described herein may serve as a dopant (or emitter), or may serve as an auxiliary dopant.
[0488] An amount (by weight) of the dopant (or emitter) in the emission layer may be in a range of about 0.01 parts by weight to about 15 parts by weight, based on 100 parts by weight of the host.
[0489] In an embodiment, the heterocyclic compound may be included in the emission layer. An amount (by weight) of the heterocyclic compound in the emission layer may be in a range of about 0.01 parts by weight to about 30 parts by weight, based on 100 parts by weight of the emission layer. For example, the amount of the heterocyclic compound in the emission layer may be in a range of about 0.1 parts by weight to about 20 parts by weight, based on 100 parts by weight of the emission layer. For example, the amount of the heterocyclic compound in the emission layer may be in a range of about 0.1 parts by weight to about 15 parts by weight, based on 100 parts by weight of the emission layer.
[0490] In embodiments, the emission layer may include a quantum dot.
[0491] In embodiments, the emission layer may include a delayed fluorescence material. The delayed fluorescence material may serve as a host or a as dopant in the emission layer.
[0492] A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the emission layer may be in a range of about 200 Å to about 600 Å. When the thickness of the emission layer is within any of the ranges described above, excellent luminescence characteristics may be obtained without a substantial increase in driving voltage.
[Host]
[0493] In an embodiment, the host in the emission layer may include the second compound, the third compound, or any combination thereof.
[0494] In embodiments, the host may include a compound represented by Formula 301:

[Ar.sub.301].sub.xb11—[(L.sub.301).sub.xb1—R.sub.301].sub.xb21 [Formula 301]
[0495] In Formula 301, [0496] Ar.sub.301 and L.sub.301 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-

C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0497] xb11 may be 1, 2, or 3, [0498] xb1 may be an integer from 0 to 5, [0499] R.sub.301 may be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(═O)(Q.sub.301), —S(═O).sub.2(Q.sub.301), or —P(═O)(Q.sub.301)(Q.sub.302), [0500] xb21 may be an integer from 1 to 5, and [0501] Q.sub.301 to Q.sub.303 may each independently be the same as described in connection with Q.sub.1.
[0502] In an embodiment, in Formula 301, when xb11 is 2 or more, two or more of Ar.sub.301 may be linked to each other via a single bond.
[0503] In embodiments, the host may include a compound represented by Formula 301-1, a compound represented by Formula 301-2, or any combination thereof:
##STR00101##
[0504] In Formulae 301-1 and 301-2, [0505] ring A.sub.301 to ring A.sub.304 may each independently be a C.sub.5-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0506] X.sub.301 may be O, S, N—[(L.sub.304).sub.xb4-R.sub.304], C(R.sub.304)(R.sub.305), or Si(R.sub.304)(R.sub.305), [0507] xb22 and xb23 may each independently be 0, 1, or 2, [0508] L.sub.301, xb1, and R.sub.301 may each be the same as described herein, [0509] L.sub.302 to L.sub.304 may each independently be the same as described in connection with L.sub.301, [0510] xb2 to xb4 may each independently be the same as described in connection with xb1, and [0511] R.sub.302 to R.sub.305 and R.sub.311 to R.sub.314 may each independently be the same as described in connection with R.sub.301.
[0512] In embodiments, the host may include an alkali earth metal complex, a post-transition metal complex, or any combination thereof. For example, the host may include a Be complex (e.g., Compound H55), a Mg complex, a Zn complex, or any combination thereof.
[0513] In embodiments, the host may include one of Compounds H1 to H128, 9,10-di(2-naphthyl) anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl) anthracene (MADN); 9,10-di(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4′-bis(N-carbazolyl)—1,1′-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri (carbazol-9-yl)benzene (TCP), or any combination thereof:
##STR00102## ##STR00103## ##STR00104## ##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111## ##STR00112## ##STR00113## ##STR00114## ##STR00115##
##STR00116## ##STR00117## ##STR00118## ##STR00119##
[0514] In an embodiment, the host may include a silicon-containing compound, a phosphine oxide-containing compound, or any combination thereof.
[0515] The host may have various modifications. For example, the host may include only one kind of compound, or may include two or more kinds of different compounds.
[Phosphorescent Dopant]
[0516] In an embodiment, the phosphorescent dopant may include at least one transition metal as a central metal.
[0517] The phosphorescent dopant may include a monodentate ligand, a bidentate ligand, a tridentate ligand, a tetradentate ligand, a pentadentate ligand, a hexadentate ligand, or any combination thereof.
[0518] The phosphorescent dopant may be electrically neutral.

[0519] In an embodiment, the phosphorescent dopant may include an organometallic compound represented by Formula 401:

M(L.sub.401).sub.xc1(L.sub.402).sub.xc2   [Formula 401]
##STR00120##
[0520] In Formulae 401 and 402, [0521] M may be a transition metal (e.g., iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), gold (Au), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), rhenium (Re), or thulium (Tm)), [0522] L.sub.401 may be a ligand represented by Formula 402, and xc1 may be 1, 2, or 3, wherein when xc1 is 2 or more, two or more of L.sub.401 may be identical to or different from each other, [0523] L.sub.402 may be an organic ligand, and xc2 may be 0, 1, 2, 3, or 4, wherein when xc2 is 2 or more, two or more of L.sub.402 may be identical to or different from each other, [0524] X.sub.401 and X.sub.402 may each independently be nitrogen or carbon, [0525] ring A.sub.401 and ring A.sub.402 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, [0526] T.sub.401 may be a single bond, *—O—**, *—S—**, *—C(═O)—**, *—N(Q.sub.411)—**, *—C(Q.sub.411)(Q.sub.412)—**, *—C(Q.sub.411)═C(Q.sub.412)—**, *—C(Q.sub.411)=**, or *═C═*′, [0527] X.sub.403 and X.sub.404 may each independently be a chemical bond (e.g., a covalent bond or a coordinate bond), O, S, N(Q.sub.413), B(Q.sub.413), P(Q.sub.413), C(Q.sub.413)(Q.sub.414), or Si(Q.sub.413)(Q.sub.414), [0528] Q.sub.411 to Q.sub.414 may each independently be the same as described in connection with Q.sub.1, [0529] R.sub.401 and R.sub.402 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(═O)(Q.sub.401), —S(═O).sub.2(Q.sub.401), or —P(═O)(Q.sub.401)(Q.sub.402), [0530] Q.sub.401 to Q.sub.403 may each independently be the same as described in connection with Q.sub.1, [0531] xc11 and xc12 may each independently be an integer from 0 to 10, and * and *′ in Formula 402 each indicate a binding site to M in Formula 401.
[0532] In an embodiment, in Formula 402, X.sub.401 may be nitrogen and X.sub.402 may be carbon, or X.sub.401 and X.sub.402 may each be nitrogen.
[0533] In embodiments, in Formula 401, when xc1 is 2 or more, two of ring A.sub.401 among two or more of L.sub.401 may optionally be linked to each other via T.sub.402, which is a linking group, and two of ring A.sub.402 among two or more of L.sub.401 may optionally be linked to each other via T.sub.403, which is a linking group (see Compounds PD1 to PD4 and PD7). T.sub.402 and T.sub.403 may each independently be the same as described in connection with T.sub.401.
[0534] In Formula 401, L.sub.402 may be an organic ligand. In an embodiment, L.sub.402 may include a halogen group, a diketone group (e.g., an acetylacetonate group), a carboxylic acid group (e.g., a picolinate group), —C(═O), an isonitrile group, a-CN group, a phosphorus group (e.g., a phosphine group, a phosphite group, etc.), or any combination thereof.
[0535] In an embodiment, the phosphorescent dopant may include, for example, one of Compounds PD1 to PD39, or any combination thereof:
##STR00121## ##STR00122## ##STR00123## ##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129##
[Fluorescent Dopant]
[0536] In an embodiment, when the emission layer includes the heterocyclic compound as described herein and the heterocyclic compound serves as an auxiliary dopant or a sensitizer, the emission layer may further include a fluorescent dopant.

[0537] In embodiments, when the emission layer includes the heterocyclic compound as described herein and the heterocyclic compound serves as a phosphorescent or fluorescent dopant, the emission layer may further include an auxiliary dopant.
[0538] The fluorescent dopant and the auxiliary dopant (or sensitizer) may each independently include an arylamine compound, a styrylamine compound, a boron-containing compound, or any combination thereof.
[0539] In an embodiment, the fluorescent dopant and the auxiliary dopant may each independently include a compound represented by Formula 501:
##STR00130## [0540] In Formula 501, [0541] Ar.sub.501, L.sub.501 to L.sub.503, R.sub.501, and R.sub.502 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0542] xd1 to xd3 may each independently be 0, 1, 2, or 3, and [0543] xd4 may be 1, 2, 3, 4, 5, or 6.
[0544] In an embodiment, in Formula 501, Ar.sub.501 may be a condensed cyclic group (e.g., an anthracene group, a chrysene group, a pyrene group, etc.) in which three or more monocyclic groups are condensed with each other.
[0545] In embodiments, in Formula 501, xd4 may be 2.
[0546] In an embodiment, the fluorescent dopant and the auxiliary dopant may each independently include one of Compounds FD1 to FD37, DPVBi, DPAVB, or any combination thereof:
##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135## ##STR00136##
[Delayed Fluorescence Material]
[0547] In an embodiment, the emission layer may include a delayed fluorescence material.
[0548] In the specification, a delayed fluorescence material may be any compound that is capable of emitting delayed fluorescence, based on a delayed fluorescence emission mechanism.
[0549] The delayed fluorescence material may include, for example, the fourth compound as described herein.
[0550] The delayed fluorescence material included in the emission layer may serve as a host or as a dopant, depending on the types of other materials included in the emission layer.
[0551] In an embodiment, a difference between a triplet energy level (eV) of the delayed fluorescence material and a singlet energy level (eV) of the delayed fluorescence material may be in a range of about 0 eV to about 0.5 eV. When the difference between the triplet energy level (eV) of the delayed fluorescence material and the singlet energy level (eV) of the delayed fluorescence material is within the range described above, up-conversion from a triplet state to a singlet state of the delayed fluorescence material may effectively occur, and thus, the light-emitting device may have improved luminescence efficiency.
[0552] In an embodiment, the delayed fluorescence material may include: a material including at least one electron donor (e.g., a π electron-rich C.sub.3-C.sub.60 cyclic group, such as a carbazole group, etc.) and at least one electron acceptor (e.g., a sulfoxide group, a cyano group, a Ir electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group, etc.); or a material including a C.sub.8-C.sub.60 polycyclic group including two or more cyclic groups that are condensed with each other while sharing B; or the like.
[0553] In an embodiment, the delayed fluorescence material may include, for example, at least one of Compounds DF1 to DF14:
##STR00137## ##STR00138## ##STR00139## ##STR00140##
[Quantum Dot]
[0554] In an embodiment, the emission layer may include a quantum dot.
[0555] In the specification, a quantum dot may be a crystal of a semiconductor compound, and may include any material capable of emitting light of various emission wavelengths according to a size of the crystal.
[0556] A diameter of the quantum dot may be, for example, in a range of about 1 nm to about 10

nm.

[0557] The quantum dot may be synthesized by a wet chemical process, a metal organic chemical vapor deposition process, a molecular beam epitaxy process, or any process similar thereto.

[0558] The wet chemical process is a method that includes mixing a precursor material with an organic solvent and growing a quantum dot particle crystal. When the crystal grows, the organic solvent naturally serves as a dispersant coordinated on the surface of the quantum dot crystal and controls the growth of the crystal so that the growth of quantum dot particles may be controlled through a process which costs less and may be more readily performed than vapor deposition methods, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE).

[0559] The quantum dot may include a Group II-VI semiconductor compound, a Group III-V semiconductor compound, a Group III-VI semiconductor compound, a Group I-III-VI semiconductor compound, a Group IV-VI semiconductor compound, a Group IV element or compound, or any combination thereof.

[0560] Examples of a Group II-VI semiconductor compound may include: a binary compound, such as CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnO, HgS, HgSe, HgTe, MgSe, MgS, or the like; a ternary compound, such as CdSeS, CdSeTe, CdSTe, ZnSeS, ZnSeTe, ZnSTe, HgSeS, HgSeTe, HgSTe, CdZnS, CdZnSe, CdZnTe, CdHgS, CdHgSe, CdHgTe, HgZnS, HgZnSe, HgZnTe, MgZnSe, MgZnS, or the like; a quaternary compound, such as CdZnSeS, CdZnSeTe, CdZnSTe, CdHgSeS, CdHgSeTe, CdHgSTe, HgZnSeS, HgZnSeTe, HgZnSTe, or the like; and any combination thereof.

[0561] Examples of a Group III-V semiconductor compound may include: a binary compound, such as GaN, GaP, GaAs, GaSb, AlN, AIP, AIAs, AISb, InN, InP, InAs, InSb, or the like; a ternary compound, such as GaNP, GaNAs, GaNSb, GaPAs, GaPSb, AlNP, AlNAs, AlNSb, AIPAS, AIPSb, InGaP, InNP, InAIP, InNAs, InNSb, InPAs, InPSb, or the like; a quaternary compound, such as GaAINP, GaAINAs, GaAINSb, GaAIPAS, GaAIPSb, GalnNP, GalnNAs, GalnNSb, GalnPAs, GalnPSb, InAINP, InAINAs, InAINSb, InAIPAs, InAIPSb, or the like; and any combination thereof. In an embodiment, a Group III-V semiconductor compound may further include a Group II element. Examples of a Group III-V semiconductor compound further including a Group II element may include InZnP, InGaZnP, InAIZnP, and the like.

[0562] Examples of a Group III-VI semiconductor compound may include: a binary compound, such as GaS, Ga.sub.Se, Ga.sub.2Se.sub.3, GaTe, InS, InSe, In.sub.2S.sub.3, In.sub.2Se.sub.3, InTe, or the like; a ternary compound, such as InGaS.sub.3, InGaSe.sub.3, or the like; and any combination thereof.

[0563] Examples of a Group I-III-VI semiconductor compound may include: a ternary compound, such as AglnS, AglnS.sub.2, CulnS, CulnS.sub.2, CuGaO.sub.2, AgGaO.sub.2, AgAIO.sub.2, or the like; and any combination thereof.

[0564] Examples of a Group IV-VI semiconductor compound may include: a binary compound, such as SnS, SnSe, SnTe, PbS, PbSe, PbTe, or the like; a ternary compound, such as SnSeS, SnSeTe, SnSTe, PbSeS, PbSeTe, PbSTe, SnPbS, SnPbSe, SnPbTe, or the like; a quaternary compound, such as SnPbSSe, SnPbSeTe, SnPbSTe, or the like; and any combination thereof.

[0565] Examples of a Group IV element or compound may include: a single element material, such as Si, Ge, or the like; a binary compound, such as SiC, SiGe, or the like; and any combination thereof.

[0566] Each element included in a compound, such as a binary compound, a ternary compound, or a quaternary compound, may be present in a particle at a uniform concentration or at a non-uniform concentration.

[0567] In an embodiment, the quantum dot may have a single structure in which the concentration of each element in the quantum dot is uniform, or the quantum dot may have a core-shell structure. For example, a material included in the core and a material included in the shell may be different

from each other.
[0568] The shell of the quantum dot may serve as a protective layer that prevents chemical degeneration of the core to maintain semiconductor characteristics, and/or may serve as a charging layer that imparts electrophoretic characteristics to the quantum dot. The shell may be single-layered or multilayered. An interface between the core and the shell may have a concentration gradient in which the concentration of a material that is present in the shell decreases toward the core.
[0569] Examples of a shell of a quantum dot may include a metal oxide, a metalloid oxide, a non-metal oxide, a semiconductor compound, or any combination thereof. Examples of a metal oxide, a metalloid oxide, or a non-metal oxide may include: a binary compound, such as SiO.sub.2, Al.sub.2O.sub.3, TiO.sub.2, ZnO, MnO, Mn.sub.2O.sub.3, Mn.sub.3O.sub.4, CuO, FeO, Fe.sub.2O.sub.3, Fe.sub.3O.sub.4, CoO, Co.sub.3O.sub.4, NiO, or the like; a ternary compound, such as MgAl.sub.2O.sub.4, CoFe.sub.2O.sub.4, NiFe.sub.2O.sub.4, CoMn.sub.2O.sub.4, or the like; and any combination thereof.
[0570] Examples of a semiconductor compound may include, as described herein, a Group II-VI semiconductor compound, a Group III-V semiconductor compound, a Group III-VI semiconductor compound, a Group I-III-VI semiconductor compound, a Group IV-VI semiconductor compound, and any combination thereof. For example, a semiconductor compound may include CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnSeS, ZnTeS, GaAs, GaP, GaSb, HgS, HgSe, HgTe, InAs, InP, InGaP, InSb, AlAs, AIP, AISb, or any combination thereof.
[0571] The quantum dot may have a full width at half maximum (FWHM) of an emission wavelength spectrum less than or equal to about 45 nm. For example, the quantum dot may have a FWHM of an emission wavelength spectrum less than or equal to about 40 nm. For example, the quantum dot may have a FWHM of an emission wavelength spectrum less than or equal to about 30 nm. When the FWHM of the quantum dot is within any of these ranges, the quantum dot may have improved color purity or improved color reproducibility. Light emitted through the quantum dot may be emitted in all directions, so that a wide viewing angle may be improved.
[0572] In an embodiment, the quantum dot may be in the form of a spherical particle, a pyramidal particle, a multi-arm particle, a cubic nanoparticle, a nanotube particle, a nanowire particle, a nanofiber particle, a nanoplate particle, or the like.
[0573] Since an energy band gap may be adjusted by controlling a size of the quantum dot, light having various wavelength bands may be obtained from a quantum dot emission layer. Accordingly, by using quantum dots of different sizes, a light-emitting device that emits light of various wavelengths may be implemented. In an embodiment, the size of the quantum dot may be selected to emit red light, green light, and/or blue light. In an embodiment, the size of the quantum dot may be configured to emit white light by a combination of light of various colors.
[Electron Transport Region in Interlayer **130**]
[0574] The electron transport region may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer including different materials, or a structure including multiple layers including different materials.
[0575] The electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.
[0576] In an embodiment, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein the layers of each structure may be stacked from the emission layer in its respective stated order, but the structure of the electron transport region is not limited thereto.
[0577] The electron transport region (e.g., a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free

compound including at least one π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group.

[0578] In an embodiment, the electron transport region may include a compound represented by Formula 601:

[Ar.sub.601].sub.xe11—[(L.sub.601).sub.xe1—R.sub.601].sub.xe21 [Formula 601]

In Formula 601,

[0579] Ar.sub.601 and L.sub.601 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, [0580] xe11 may be 1, 2, or 3, [0581] xe1 may be 0, 1, 2, 3, 4, or 5, [0582] R.sub.601 may be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.601)(Q.sub.602)(Q.sub.603), —C(═O)(Q.sub.601), —S(═O).sub.2(Q.sub.601), or —P(═O)(Q.sub.601)(Q.sub.602), [0583] Q.sub.601 to Q.sub.603 may each independently be the same as described in connection with Q.sub.1, [0584] xe21 may be 1, 2, 3, 4, or 5, and [0585] at least one of Ar.sub.601, L.sub.601, and R.sub.601 may each independently be a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

[0586] In an embodiment, in Formula 601, when xe11 is 2 or more, two or more of Ar.sub.601 may be linked to each other via a single bond.

[0587] In embodiments, in Formula 601, Ar.sub.601 may be a substituted or unsubstituted anthracene group.

[0588] In embodiments, the electron transport region may include a compound represented by Formula 601-1:

##STR00141##

[0589] In Formula 601-1, [0590] X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or C(R.sub.615), X.sub.616 may be N or C(R.sub.616), and at least one of X.sub.614 to X.sub.616 may each be N, [0591] L.sub.611 to L.sub.613 may each independently be the same as described in connection with L.sub.601, [0592] xe611 to xe613 may each independently be the same as described in connection with xe1, [0593] R.sub.611 to R.sub.613 may each independently be the same as described in connection with R.sub.601, and [0594] R.sub.614 to R.sub.616 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

[0595] In an embodiment, in Formulae 601 and 601-1, xe1 and xe611 to xe613 may each independently be 0, 1, or 2.

[0596] The electron transport region may include one of Compounds ET1 to ET45, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq.sub.3, BAlq, TAZ, NTAZ, or any combination thereof:

##STR00142## ##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148## ##STR00149## ##STR00150## ##STR00151## ##STR00152## ##STR00153## ##STR00154## ##STR00155## ##STR00156##

[0597] A thickness of the electron transport region may be in a range of about 100 Å to about 5,000 Å. For example, the thickness of the electron transport region may be in a range of about 160 Å to about 4,000 Å. When the electron transport region includes a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, or any combination thereof, a thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 20 Å to about 1,000 Å, and a thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the buffer layer, the hole

blocking layer, or the electron control layer may each independently be in a range of about 30 Å to about 300 Å. For example, the thickness of the electron transport layer may be in a range of about 150 Å to about 500 Å. When the thicknesses of the buffer layer, the hole blocking layer, the electron control layer, the electron transport layer, and/or the electron transport region are within the ranges described above, satisfactory electron transporting characteristics may be obtained without a substantial increase in driving voltage.
[0598] The electron transport region (e.g., an electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.
[0599] The metal-containing material may include an alkali metal complex, an alkaline earth metal complex, or any combination thereof. A metal ion of the alkali metal complex may be a Li ion, a Na ion, a K ion, a Rb ion, or a Cs ion, and a metal ion of the alkaline earth metal complex may be a Be ion, a Mg ion, a Ca ion, a Sr ion, or a Ba ion.
[0600] A ligand coordinated with a metal ion of an alkali metal complex or an alkaline earth-metal complex may each independently include a hydroxyquinoline, a hydroxyisoquinoline, a hydroxybenzoquinoline, a hydroxyacridine, a hydroxyphenanthridine, a hydroxyphenyloxazole, a hydroxyphenylthiazole, a hydroxyphenyloxadiazole, a hydroxyphenylthiadiazole, a hydroxyphenylpyridine, a hydroxyphenylbenzimidazole, a hydroxyphenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.
[0601] In an embodiment, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (LiQ) or Compound ET-D2:
##STR00157##
[0602] The electron transport region may include an electron injection layer that facilitates the injection of electrons from the second electrode **150**. The electron injection layer may contact (e.g., directly contact) the second electrode **150**.
[0603] The electron injection layer may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer including different materials, or a structure including multiple layers including different materials.
[0604] In an embodiment, the electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof.
[0605] The alkali metal may include Li, Na, K, Rb, Cs, or any combination thereof. The alkaline earth metal may include Mg, Ca, Sr, Ba, or any combination thereof. The rare earth metal may include Sc, Y, Ce, Tb, Yb, Gd, or any combination thereof.
[0606] The alkali metal-containing compound, the alkaline earth metal-containing compound, and the rare earth metal-containing compound may include oxides, halides (e.g., fluorides, chlorides, bromides, iodides, etc.), or tellurides of the alkali metal, the alkaline earth metal, and the rare earth metal, or any combination thereof.
[0607] The alkali metal-containing compound may include: an alkali metal oxide, such as Li_2O, Cs_2O, K_2O, or the like; an alkali metal halide, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, KI, RbI, or the like; or any combination thereof. The alkaline earth metal-containing compound may include an alkaline earth metal oxide, such as BaO, SrO, CaO, $Ba_xSr_{1-x}O$ (wherein x is a real number satisfying $0<x<1$), $Ba_xCa_{1-x}O$ (wherein x is a real number satisfying $0<x<1$), or the like. The rare earth metal-containing compound may include YbF_3, ScF_3, SC_2O_3, Y_2O_3, Ce_2O_3, GdF_3, TbF_3, YbI_3, ScI_3, TbI_3, or any combination thereof. In an embodiment, the rare earth metal-containing compound may include a lanthanide metal telluride. Examples of a lanthanide metal telluride may include LaTe, CeTe, PrTe, NdTe, PmTe, SmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, LuTe, La_2Te_3, Ce_2Te_3, Pr_2Te_3, Nd_2Te_3,

Pm_2Te_3, Sm_2Te_3, Eu_2Te_3, Gd_2Te_3, Tb_2Te_3, Dy_2Te_3, Ho_2Te_3, Er_2Te_3, Tm_2Te_3, Yb_2Te_3, Lu_2Te_3, and the like.

[0608] The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include: an alkali metal ion, an alkaline earth metal ion, or a rare earth metal ion; and a ligand bonded to the metal ion (for example, a hydroxyquinoline, a hydroxyisoquinoline, a hydroxybenzoquinoline, a hydroxyacridine, a hydroxyphenanthridine, a hydroxyphenyloxazole, a hydroxyphenylthiazole, hydroxyphenyloxadiazole, a hydroxyphenylthiadiazole, a hydroxyphenylpyridine, a hydroxyphenylbenzimidazole, a hydroxyphenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof).

[0609] In an embodiment, the electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof, as described above. In embodiments, the electron injection layer may further include an organic material (e.g., a compound represented by Formula 601).

[0610] In an embodiment, the electron injection layer may consist of an alkali metal-containing compound (e.g., an alkali metal halide); or the electron injection layer may consist of an alkali metal-containing compound (e.g., an alkali metal halide), and an alkali metal, an alkaline earth metal, a rare earth metal, or any combination thereof. In an embodiment, the electron injection layer may be a KI: Yb co-deposited layer, an Rbl: Yb co-deposited layer, a LiF: Yb co-deposited layer, or the like.

[0611] When the electron injection layer further includes an organic material, the alkali metal, the alkaline earth metal, the rare earth metal, the alkali metal-containing compound, the alkaline earth metal-containing compound, the rare earth metal-containing compound, the alkali metal complex, the alkaline earth-metal complex, the rare earth metal complex, or any combination thereof may be uniformly or non-uniformly dispersed in a matrix including the organic material.

[0612] A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å. For example, the thickness of the electron injection layer may be in a range of about 3 Å to about 90 Å. When the thickness of the electron injection layer is within any of the ranges described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

[Second Electrode **150**]

[0613] The second electrode **150** may be arranged on the interlayer **130** having a structure as described above. The second electrode **150** may be a cathode, which is an electron injection electrode. When the second electrode **150** is a cathode, a material for forming the second electrode may include a material having a low-work function, such as a metal, an alloy, an electrically conductive compound, or any combination thereof.

[0614] The second electrode **150** may include lithium (Li), silver (Ag), magnesium (Mg), aluminum (AI), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ytterbium (Yb), silver-ytterbium (Ag—Yb), ITO, IZO, or any combination thereof. The second electrode **150** may be a transmissive electrode, a transflective electrode, or a reflective electrode.

[0615] The second electrode **150** may have a single-layer structure or a multi-layer structure.

[Capping Layer]

[0616] The light-emitting device **10** may include a first capping layer outside the first electrode **110**, and/or a second capping layer outside the second electrode **150**. In embodiments, the light-emitting device **10** may have a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, and the second electrode **150** are stacked in this stated order, a structure in which the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are stacked in this stated order, or a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are stacked in this stated order.

[0617] Light generated in the emission layer of the interlayer **130** of the light-emitting device **10** may be extracted through the first electrode **110**, which may be a transflective electrode or a transmissive electrode, and through the first capping layer to the outside. Light generated in the emission layer of the interlayer **130** of the light-emitting device **10** may be extracted through the second electrode **150**, which may be a transflective electrode or a transmissive electrode, and through the second capping layer to the outside.

[0618] The first capping layer and the second capping layer may each increase external emission efficiency according to the principle of constructive interference. Accordingly, the light-extraction efficiency of the light-emitting device **10** may be increased, so that the luminescence efficiency of the light-emitting device **10** may be improved.

[0619] The first capping layer and the second capping layer may each include a material having a refractive index greater than or equal to about 1.6 (with respect to a wavelength of about 589 nm).

[0620] The first capping layer and the second capping layer may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or an organic-inorganic composite capping layer including an organic material and an inorganic material.

[0621] At least one of the first capping layer and the second capping layer may each independently include a carbocyclic compound, a heterocyclic compound, an amine group-containing compound, a porphine derivative, a phthalocyanine derivative, a naphthalocyanine derivative, an alkali metal complex, an alkaline earth metal complex, or any combination thereof. The carbocyclic compound, the heterocyclic compound, and the amine group-containing compound may optionally be substituted with a substituent including O, N, S, Se, Si, F, Cl, Br, I, or any combination thereof.

[0622] In an embodiment, at least one of the first capping layer and the second capping layer may each independently include an amine group-containing compound.

[0623] In an embodiment, at least one of the first capping layer and the second capping layer may each independently include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof.

[0624] In embodiments, at least one of the first capping layer and the second capping layer may each independently include one of Compounds HT28 to HT33, one of Compounds CP1 to CP6, β-NPB, or any combination thereof:

##STR00158## ##STR00159##

[Film]

[0625] The heterocyclic compound may be included in various films. Accordingly, another embodiment provides a film including the heterocyclic compound. The film may be, for example, an optical member (or a light control means)(e.g., a color filter, a color conversion member, a capping layer, a light extraction efficiency enhancement layer, a selective light absorbing layer, a polarizing layer, a quantum dot-containing layer, etc.), a light blocking member (e.g., a light reflective layer, a light absorbing layer, etc.), a protective member (e.g., an insulating layer, a dielectric layer, etc.), or the like.

[Electronic Apparatus]

[0626] The light-emitting device may be included in various electronic apparatuses. For example, an electronic apparatus including the light-emitting device may be a light-emitting apparatus, an authentication apparatus, or the like.

[0627] The electronic apparatus (e.g., a light-emitting apparatus) may further include, in addition to the light-emitting device, a color filter, a color conversion layer, or a color filter and a color conversion layer. The color filter and/or the color conversion layer may be arranged in at least one traveling direction of light emitted from the light-emitting device. For example, light emitted from the light-emitting device may be blue light, green light, or white light. The light-emitting device may be the same as described herein. In an embodiment, the color conversion layer may include a quantum dot.

[0628] The electronic apparatus may include a substrate. The substrate may include subpixels, the color filter may include color filter areas respectively corresponding to the subpixels, and the color conversion layer may include color conversion areas respectively corresponding to the subpixels.

[0629] A pixel-defining film may be arranged between the subpixels to define each subpixel.

[0630] The color filter may further include color filter areas and light-shielding patterns arranged between the color filter areas, and the color conversion layer may further include color conversion areas and light-shielding patterns arranged between the color conversion areas.

[0631] The color filter areas (or the color conversion areas) may include a first area emitting first color light, a second area emitting second color light, and/or a third area emitting third color light, wherein the first color light, the second color light, and/or the third color light may have different maximum emission wavelengths from one another. In an embodiment, the first color light may be red light, the second color light may be green light, and the third color light may be blue light. In an embodiment, the color filter areas (or the color conversion areas) may include quantum dots. In an embodiment, the first area may include a red quantum dot, the second area may include a green quantum dot, and the third area may not include a quantum dot. The quantum dot may be the same as described herein. The first area, the second area, and/or the third area may each further include a scatterer.

[0632] In an embodiment, the light-emitting device may emit first light, the first area may absorb the first light to emit first-first color light, the second area may absorb the first light to emit second-first color light, and the third area may absorb the first light to emit third-first color light. The first-first color light, the second-first color light, and the third-first color light may have different maximum emission wavelengths. In an embodiment, the first light may be blue light, the first-first color light may be red light, the second-first color light may be green light, and the third-first color light may be blue light.

[0633] The electronic apparatus may further include a thin-film transistor, in addition to the light-emitting device as described above. The thin-film transistor may include a source electrode, a drain electrode, and an active layer, wherein any one of the source electrode and the drain electrode may be electrically connected to any one of the first electrode and the second electrode of the light-emitting device.

[0634] The thin-film transistor may further include a gate electrode, a gate insulating film, and the like.

[0635] The active layer may include crystalline silicon, amorphous silicon, an organic semiconductor, an oxide semiconductor, and the like.

[0636] The electronic apparatus may further include a sealing portion for sealing the light-emitting device. The sealing portion may be arranged between the color filter and/or the color conversion layer and the light-emitting device. The sealing portion may allow light from the light-emitting device to be extracted to the outside, and may prevent ambient air and moisture from penetrating into the light-emitting device. The sealing portion may be a sealing substrate that includes a transparent glass substrate or a plastic substrate. The sealing portion may be a thin-film encapsulation layer that includes at least one of an organic layer and/or an inorganic layer. When the sealing portion is a thin-film encapsulation layer, the electronic apparatus may be flexible.

[0637] Various functional layers may be further included on the sealing portion, in addition to the color filter and/or the color conversion layer, according to a use of the electronic apparatus. Examples of a functional layer may include a touch screen layer, a polarizing layer, and the like. The touch screen layer may be a pressure-sensitive touch screen layer, a capacitive touch screen layer, or an infrared touch screen layer. The authentication apparatus may be, for example, a biometric authentication apparatus that authenticates an individual by using biometric information of a living body (e.g., fingertips, pupils, etc.).

[0638] The authentication apparatus may further include, in addition to the light-emitting device as described above, a biometric information collector.

[0639] The electronic apparatus may be applied to various displays, light sources, lighting, personal computers (e.g., a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (e.g., electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram displays, ultrasonic diagnostic devices, or endoscope displays), fish finders, various measuring instruments, meters (e.g., meters for a vehicle, an aircraft, and a vessel), projectors, and the like.

[Electronic Equipment]

[0640] The light-emitting device may be included in various electronic equipment.

[0641] In embodiments, an electronic equipment including the light-emitting device may be a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.

[0642] The light-emitting device may have excellent luminescence efficiency and long lifespan, and thus, the electronic equipment including the light-emitting device may have characteristics such as high luminance, high resolution, and low power consumption.

[Descriptions of FIGS. **2** and **3**]

[0643] FIG. **2** is a schematic cross-sectional view of an electronic apparatus (e.g., a light-emitting apparatus) according to an embodiment.

[0644] The electronic apparatus of FIG. **2** may include a substrate **100**, a thin-film transistor (TFT), a light-emitting device, and an encapsulation portion **300** that seals the light-emitting device.

[0645] The substrate **100** may be a flexible substrate, a glass substrate, or a metal substrate. A buffer layer **210** may be arranged on the substrate **100**. The buffer layer **210** may prevent penetration of impurities through the substrate **100**, and may provide a flat surface on the substrate **100**.

[0646] A TFT may be arranged on the buffer layer **210**. The TFT may include an active layer **220**, a gate electrode **240**, a source electrode **260**, and a drain electrode **270**.

[0647] The active layer **220** may include an inorganic semiconductor, such as silicon or polysilicon, an organic semiconductor, or an oxide semiconductor, and may include a source region, a drain region, and a channel region.

[0648] A gate insulating film **230** for insulating the active layer **220** from the gate electrode **240** may be arranged on the active layer **220**, and the gate electrode **240** may be arranged on the gate insulating film **230**.

[0649] An interlayer insulating film **250** may be arranged on the gate electrode **240**. The interlayer insulating film **250** may be arranged between the gate electrode **240** and the source electrode **260** to insulate the gate electrode **240** from the source electrode **260** and between the gate electrode **240** and the drain electrode **270** to insulate the gate electrode **240** from the drain electrode **270**.

[0650] The source electrode **260** and the drain electrode **270** may be arranged on the interlayer insulating film **250**. The interlayer insulating film **250** and the gate insulating film **230** may be formed to expose a source region and a drain region of the active layer **220**, and the source electrode **260** and the drain electrode **270** may respectively contact the exposed portions of the source region and the drain region of the active layer **220**.

[0651] The TFT may be electrically connected to a light-emitting device to drive the light-emitting device, and may be covered and protected by a passivation layer **280**. The passivation layer **280** may include an inorganic insulating film, an organic insulating film, or any combination thereof. A light-emitting device may be provided on the passivation layer **280**. The light-emitting device may include the first electrode **110**, the interlayer **130**, and the second electrode **150**.
[0652] The first electrode **110** may be arranged on the passivation layer **280**. The passivation layer **280** may not completely cover the drain electrode **270** and may expose a portion of the drain electrode **270**. The first electrode **110** may be connected (for example, electrically connected) to the exposed portion of the drain electrode **270**.
[0653] A pixel-defining film **290** including an insulating material may be arranged on the first electrode **110**. The pixel-defining film **290** may expose a region of the first electrode **110**, and the interlayer **130** may be formed on the exposed region of the first electrode **110**. The pixel-defining film **290** may be a polyimide-based organic film or a polyacrylic organic film. Although not shown in FIG. **2**, at least some layers of the interlayer **130** may extend beyond the upper portion of the pixel-defining film **290** to be provided in the form of a common layer.
[0654] The second electrode **150** may be arranged on the interlayer **130**, and a second capping layer **170** may be further included on the second electrode **150**. The second capping layer **170** may be formed to cover the second electrode **150**.
[0655] The encapsulation portion **300** may be arranged on the second capping layer **170**. The encapsulation portion **300** may be arranged on a light-emitting device to protect the light-emitting device from moisture and/or oxygen. The encapsulation portion **300** may include: an inorganic film including silicon nitride (SiNx), silicon oxide (SiOx), indium tin oxide, indium zinc oxide, or any combination thereof; an organic film including polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyimide, polyethylene sulfonate, polyoxymethylene, polyarylate, hexamethyldisiloxane, an acrylic resin (e.g., polymethyl methacrylate, polyacrylic acid, etc.), an epoxy-based resin (e.g., aliphatic glycidyl ether (AGE), etc.), or any combination thereof; or any combination of the inorganic film and the organic film.
[0656] FIG. **3** is a schematic cross-sectional view of an electronic apparatus according to another embodiment.
[0657] The electronic apparatus of FIG. **3** may differ from the electronic apparatus of FIG. **2**, at least in that a light-shielding pattern **500** and a functional region **400** are further included on the encapsulation portion **300**. The functional region **400** may be a color filter area, a color conversion area, or a combination of the color filter area and the color conversion area. In an embodiment, the light-emitting device included in the electronic apparatus of FIG. **3** may be a tandem light-emitting device. [Description of FIG. **4**]
[0658] FIG. **4** is a schematic perspective view of electronic equipment **1** including a light-emitting device according to an embodiment.
[0659] The electronic equipment **1**, which may be an apparatus that displays a moving image or still image, may not only be a portable electronic equipment, such as a mobile phone, a smartphone, a tablet computer, a mobile communication terminal, an electronic notebook, an electronic book, a portable multimedia player (PMP), a navigation device, or an ultra-mobile PC (UMPC), but may also be various products, such as a television, a laptop, a monitor, a billboard, or an Internet of things (IoT). The electronic equipment **1** may be any product as described above or a part thereof.
[0660] In an embodiment, the electronic equipment **1** may be a wearable device, such as a smart watch, a watch phone, a glasses-type display, or a head mounted display (HMD), or a part of the wearable device. However, embodiments are not limited thereto.
[0661] Examples of the electronic equipment **1** may include a dashboard of a vehicle, a center information display (CID) arranged on a center fascia or dashboard of a vehicle, a room mirror display that replaces a side mirror of a vehicle, an entertainment display arranged for a rear seat of

a vehicle or arranged on the back of a front seat, a head-up display (HUD) installed at the front of a vehicle or projected on a front window glass, or a computer generated hologram augmented reality head up display (CGH AR HUD). FIG. **4** illustrates an embodiment in which the electronic equipment **1** is a smartphone, for convenience of explanation.

[0662] The electronic equipment **1** may include a display area DA and a non-display area NDA outside the display area DA. A display apparatus may implement an image through a two-dimensional array of pixels that are arranged in the display area DA.

[0663] The non-display area NDA may be an area that does not display an image, and may surround (for example, entirely surround) the display area DA. A driver for providing electrical signals or power to display devices arranged on the display area DA may be arranged in the non-display area NDA. A pad, which is an area to which an electronic element or a printed circuit board may be electrically connected, may be arranged in the non-display area NDA.

[0664] In the electronic equipment **1**, a length in an x-axis direction and a length in a y-axis direction may be different from each other. In an embodiment, as shown in FIG. **4**, a length in the x-axis direction may be less than a length in the y-axis direction. In embodiments, a length in the x-axis direction may be the same as a length in the y-axis direction. In embodiments, a length in the x-axis direction may be longer than a length in the y-axis direction.

[Descriptions of FIGS. **5** and **6**A to **6**C]

[0665] FIG. **5** is a schematic perspective view of an exterior of a vehicle **1000** as an electronic equipment including the light-emitting device according to an embodiment. FIGS. **6**A to **6**C are each a schematic diagram of an interior of the vehicle **1000** according to embodiments.

[0666] Referring to FIGS. **5**, **6**A, **6**B, and **6**C, embodiments of a vehicle **1000** may include various apparatuses for moving a subject to be transported, such as a person, an object, or an animal, from a departure point to a destination point. Examples of a vehicle **1000** may include a vehicle traveling on a road or track, a vessel moving over the sea or river, an airplane flying in the sky using the action of air, and the like.

[0667] The vehicle **1000** may travel on a road or a track. The vehicle **1000** may move in a selectable direction according to the rotation of at least one wheel. Examples of a vehicle **1000** may include a three-wheeled or four-wheeled vehicle, a construction machine, a two-wheeled vehicle, a prime mover device, a bicycle, and a train running on a track.

[0668] The vehicle **1000** may include a vehicle body having an interior and an exterior, and a chassis that is a portion excluding the vehicle body in which mechanical apparatuses necessary for driving are installed. The exterior of the vehicle body may include a front panel, a bonnet, a roof panel, a rear panel, a trunk, a pillar provided at a boundary between doors, and the like. The chassis of the vehicle **1000** may include a power generating device, a power transmitting device, a driving device, a steering device, a braking device, a suspension device, a transmission device, a fuel device, front and rear wheels, left and right wheels, and the like.

[0669] The vehicle **1000** may include a side window glass **1100**, a front window glass **1200**, a side mirror **1300**, a cluster **1400**, a center fascia **1500**, a passenger seat dashboard **1600**, and a display apparatus **2**.

[0670] The side window glass **1100** and the front window glass **1200** may be partitioned by a pillar arranged between the side window glass **1100** and the front window glass **1200**.

[0671] The side window glass **1100** may be installed on a side of the vehicle **1000**. In an embodiment, the side window glass **1100** may be installed on a door of the vehicle **1000**. Multiple side window glasses **1100** may be provided and may face each other. In an embodiment, the side window glass **1100** may include a first side window glass **1110** and a second side window glass **1120**. In an embodiment, the first side window glass **1110** may be arranged adjacent to the cluster **1400**, and the second side window glass **1120** may be arranged adjacent to the passenger seat dashboard **1600**.

[0672] In an embodiment, the side window glasses **1100** may be spaced apart from each other in the x-direction or the −x-direction. In an embodiment, the first side window glass **1110** and the second side window glass **1120** may be spaced apart from each other in the x-direction or the −x-direction. For example, a virtual straight line L connecting the side window glasses **1100** may extend in the x-direction or the −x-direction. For example, a virtual straight line L connecting the first side window glass **1110** and the second side window glass **1120** to each other may extend in the x-direction or the −x-direction.

[0673] The front window glass **1200** may be installed in the front of the vehicle **1000**. The front window glass **1200** may be arranged between the side window glasses **1100** facing each other.

[0674] The side mirror **1300** may provide a rear view of the vehicle **1000**. The side mirror **1300** may be installed on the exterior of the vehicle body. In an embodiment, multiple side mirrors **1300** may be provided. For example, one of the side mirrors **1300** may be arranged outside the first side window glass **1110**, and another of the side mirrors **1300** may be arranged outside the second side window glass **1120**.

[0675] The cluster **1400** may be arranged in front of the steering wheel. The cluster **1400** may include a tachometer, a speedometer, a coolant thermometer, a fuel gauge, a turn signal indicator, a high beam indicator, a warning light, a seat belt warning light, an odometer, a tachograph, an automatic shift selector indicator light, a door open warning light, an engine oil warning light, and/or a low fuel warning light.

[0676] The center fascia **1500** may include a control panel on which buttons for adjusting an audio device, an air conditioning device, and a seat heater are arranged. The center fascia **1500** may be arranged on a side of the cluster **1400**.

[0677] The passenger seat dashboard **1600** may be spaced apart from the cluster **1400**, and the center fascia **1500** may be arranged between the cluster **1400** and the passenger seat dashboard **1600**. In an embodiment, the cluster **1400** may be arranged to correspond to a driver seat (not shown), and the passenger seat dashboard **1600** may be arranged to correspond to a passenger seat (not shown). In an embodiment, the cluster **1400** may be adjacent to the first side window glass **1110**, and the passenger seat dashboard **1600** may be adjacent to the second side window glass **1120**.

[0678] In an embodiment, the display apparatus **2** may include a display panel **3**, and the display panel **3** may display an image. The display apparatus **2** may be arranged inside the vehicle **1000**. In an embodiment, the display apparatus **2** may be arranged between the side window glasses **1100** facing each other. The display apparatus **2** may be arranged on at least one of the cluster **1400**, the center fascia **1500**, and the passenger seat dashboard **1600**.

[0679] The display apparatus **2** may include an organic light-emitting display apparatus, an inorganic light-emitting display apparatus, a quantum dot display apparatus, or the like. Hereinafter, an organic light-emitting display apparatus including the light-emitting device according to an embodiment will be described as an example of the display apparatus **2**. However, various types of display apparatuses as described above may be used in embodiments.

[0680] Referring to FIG. **6**A, the display apparatus **2** may be arranged on the center fascia **1500**. In an embodiment, the display apparatus **2** may display navigation information. In an embodiment, the display apparatus **2** may display information regarding audio settings, video settings, or vehicle settings.

[0681] Referring to FIG. **6**B, the display apparatus **2** may be arranged on the cluster **1400**. In an embodiment, the cluster **1400** may display driving information and the like through the display apparatus **2**. For example, the cluster **1400** may digitally implement driving information. The digital cluster **1400** may display vehicle information and driving information as images. For example, a needle and a gauge of a tachometer and various warning lights or icons may be displayed by a digital signal.

[0682] Referring to FIG. **6**C, the display apparatus **2** may be arranged on the passenger seat dashboard **1600**. The display apparatus **2** may be embedded in the passenger seat dashboard **1600** or arranged on the passenger seat dashboard **1600**. In an embodiment, the display apparatus **2** arranged on the passenger seat dashboard **1600** may display an image related to information displayed on the cluster **1400** and/or information displayed on the center fascia **1500**. In embodiments, the display apparatus **2** arranged on the passenger seat dashboard **1600** may display information that is different from information displayed on the cluster **1400** and/or information displayed on the center fascia **1500**.

[Manufacturing Method]

[0683] The layers constituting the hole transport region, the emission layer, and the layers constituting the electron transport region may be formed in a selected region by using various methods such as vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, laser-induced thermal imaging, and the like.

[0684] When the layers constituting the hole transport region, the emission layer, and the layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature in a range of about 100° C. to about 500° C., at a vacuum degree in a range of about 10-8 torr to about 10-3 torr, and at a deposition speed in a range of about 0.01 Å/see to about 100 Å/see, depending on a material to be included in a layer to be formed and the structure of a layer to be formed.

[Definitions of Terms]

[0685] The term “C.sub.3-C.sub.60 carbocyclic group” as used herein may be a cyclic group consisting of carbon atoms as the only ring-forming atoms and having 3 to 60 carbon atoms. The term “C.sub.1-C.sub.60 heterocyclic group” as used herein may be a cyclic group that has 1 to 60 carbon atoms and further has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom. The C.sub.3-C.sub.60 carbocyclic group and the C.sub.1-C.sub.60 heterocyclic group may each be a monocyclic group consisting of one ring or a polycyclic group in which two or more rings are condensed with each other. For example, the number of ring-forming atoms in a C.sub.1-C.sub.60 heterocyclic group may be from 3 to 61.

[0686] The term “cyclic group” as used herein may be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group.

[0687] The term “π electron-rich C.sub.3-C.sub.60 cyclic group” as used herein may be a cyclic group that has 3 to 60 carbon atoms and may not include *—N=*′ as a ring-forming moiety. The term “π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group” as used herein may be a heterocyclic group that has 1 to 60 carbon atoms and may include *—N=*′ as a ring-forming moiety.

[0688] In an embodiment,

[0689] a C.sub.3-C.sub.60 carbocyclic group may be a T.sub.1 group or a group in which two or more T.sub.1 groups are condensed with each other (e.g., a cyclopentadiene group, an adamantane group, a norbornane group, a benzene group, a pentalene group, a naphthalene group, an azulene group, an indacene group, an acenaphthylene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a perylene group, a pentaphene group, a heptalene group, a naphthacene group, a picene group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, an indenophenanthrene group, or an indenoanthracene group),

[0690] a C.sub.1-C.sub.60 heterocyclic group may be a T2 group, a group in which two or more T2 groups are condensed with each other, or a group in which at least one T2 group and at least one T.sub.1 group are condensed with each other (e.g., a pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.),

[0691] a Ir electron-rich C.sub.3-C.sub.60 cyclic group may be a T.sub.1 group, a group in which two or more T.sub.1 groups are condensed with each other, a T3 group, a group in which two or more T3 groups are condensed with each other, or a group in which at least one T3 group and at least one T.sub.1 group are condensed with each other (e.g., a C.sub.3-C.sub.60 carbocyclic group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, etc.),

[0692] a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group may be a T4 group, a group in which two or more of T4 groups are condensed with each other, a group in which at least one T4 group and at least one T.sub.1 group are condensed with each other, a group in which at least one T4 group and at least one T3 group are condensed with each other, or a group in which at least one T4 group, at least one T.sub.1 group, and at least one T3 group are condensed with one another (e.g., a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.),

[0693] wherein the T.sub.1 group may be a cyclopropane group, a cyclobutane group, a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclobutene group, a cyclopentene group, a cyclopentadiene group, a cyclohexene group, a cyclohexadiene group, a cycloheptene group, an adamantane group, a norbornane (or bicyclo[2.2.1] heptane) group, a norbornene group, a bicyclo[1.1.1] pentane group, a bicyclo[2.1.1] hexane group, a bicyclo[2.2.2] octane group, or a benzene group,

[0694] the T2 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, a

borole group, a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a tetrazine group, a pyrrolidine group, an imidazolidine group, a dihydropyrrole group, a piperidine group, a tetrahydropyridine group, a dihydropyridine group, a hexahydropyrimidine group, a tetrahydropyrimidine group, a dihydropyrimidine group, a piperazine group, a tetrahydropyrazine group, a dihydropyrazine group, a tetrahydropyridazine group, or a dihydropyridazine group,
[0695] the T3 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, or a borole group, and
[0696] the T4 group may be a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a tetrazine group.
[0697] The terms “cyclic group”, “C.sub.3-C.sub.60 carbocyclic group”, “C.sub.1-C.sub.60 heterocyclic group”, “π electron-rich C.sub.3-C.sub.60 cyclic group”, and “π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group” as used herein may each be a group condensed to any cyclic group, a monovalent group, or a polyvalent group (e.g., a divalent group, a trivalent group, a tetravalent group, etc.) according to the structure of a formula for which the corresponding term is used. For example, a “benzene group” may be a benzo group, a phenyl group, a phenylene group, or the like, which may be readily understood by those of ordinary skill in the art according to the structure of a formula including the “benzene group”.
[0698] Examples of a monovalent C.sub.3-C.sub.60 carbocyclic group or a monovalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.5-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group. Examples of a divalent C.sub.3-C.sub.60 carbocyclic group or a divalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkylene group, a C.sub.1-C.sub.10 heterocycloalkylene group, a C.sub.5-C.sub.10 cycloalkenylene group, a C.sub.1-C.sub.10 heterocycloalkenylene group, a C.sub.6-C.sub.60 arylene group, a C.sub.1-C.sub.60 heteroarylene group, a divalent non-aromatic condensed polycyclic group, and a divalent non-aromatic condensed heteropolycyclic group.
[0699] The term “C.sub.1-C.sub.60 alkyl group” as used herein may be a linear or branched monovalent aliphatic hydrocarbon group that has 1 to 60 carbon atoms, and examples thereof may include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, a tert-decyl group, and the like. The term “C.sub.1-C.sub.60 alkylene group” as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.60 alkyl group.
[0700] The term “C.sub.2-C.sub.60 alkenyl group” as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon double bond in the middle or at a terminus of a C.sub.2-C.sub.60 alkyl group, and examples thereof may include an ethenyl group, a propenyl group, a butenyl group, and the like. The term “C.sub.2-C.sub.60 alkenylene group” as used herein may be a divalent group having a same structure as the C.sub.2-C.sub.60 alkenyl group.

[0701] The term "C.sub.2-C.sub.60 alkynyl group" as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon triple bond in the middle or at a terminus of a C.sub.2-C.sub.60 alkyl group, and examples thereof may include an ethynyl group, a propynyl group, and the like. The term "C.sub.2-C.sub.60 alkynylene group" as used herein may be a divalent group having a same structure as the C.sub.2-C.sub.60 alkynyl group.
[0702] The term "C.sub.1-C.sub.60 alkoxy group" as used herein may be a monovalent group represented by —O(A.sub.101)(wherein A.sub.101 may be a C.sub.1-C.sub.60 alkyl group), and examples thereof may include a methoxy group, an ethoxy group, an isopropyloxy group, and the like.
[0703] The term "C.sub.3-C.sub.10 cycloalkyl group" as used herein may be a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or bicyclo[2.2.1] heptyl group), a bicyclo[1.1.1] pentyl group, a bicyclo[2.1.1] hexyl group, a bicyclo[2.2.2] octyl group, and the like. The term "C.sub.3-C.sub.10 cycloalkylene group" as used herein may be a divalent group having a same structure as the C.sub.3-C.sub.10 cycloalkyl group.
[0704] The term "C.sub.1-C.sub.10 heterocycloalkyl group" as used herein may be a monovalent cyclic group having 1 to 10 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms, and examples thereof may include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, a tetrahydrothiophenyl group, and the like. The term "C.sub.1-C.sub.10 heterocycloalkylene group" as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.
[0705] The term "C.sub.3-C.sub.10 cycloalkenyl group" as used herein may be a monovalent cyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the cyclic structure thereof and no aromaticity, and examples thereof may include a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, and the like. The term "C.sub.3-C.sub.10 cycloalkenylene group" as used herein may be a divalent group having a same structure as the C.sub.3-C.sub.10 cycloalkenyl group.
[0706] The term "C.sub.1-C.sub.10 heterocycloalkenyl group" as used herein may be a monovalent cyclic group that has 1 to 10 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms, and having at least one carbon-carbon double bond in the cyclic structure thereof. Examples of a C.sub.1-C.sub.10 heterocycloalkenyl group may include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, a 2,3-dihydrothiophenyl group, and the like. The term "C.sub.1-C.sub.10 heterocycloalkenylene group" as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.
[0707] The term "C.sub.6-C.sub.60 aryl group" as used herein may be a monovalent group having a carbocyclic aromatic system of 6 to 60 carbon atoms, and the term "C.sub.6-C.sub.60 arylene group" as used herein may be a divalent group having a carbocyclic aromatic system of 6 to 60 carbon atoms. Examples of a C.sub.6-C.sub.60 aryl group may include a phenyl group, a pentalenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a heptalenyl group, a naphthacenyl group, a picenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, and the like. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the respective two or more rings may be condensed with each other.
[0708] The term "C.sub.1-C.sub.60 heteroaryl group" as used herein may be a monovalent group having a heterocyclic aromatic system of 1 to 60 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms. The term "C.sub.1-C.sub.60 heteroarylene group" as used herein may be a divalent group having a heterocyclic aromatic system

of 1 to 60 carbon atoms, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms. Examples of a C.sub.1-C.sub.60 heteroaryl group may include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, a benzoquinolinyl group, an isoquinolinyl group, a benzoisoquinolinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthrolinyl group, a phthalazinyl group, a naphthyridinyl group, and the like. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the respective two or more rings may be condensed with each other.
[0709] The term “monovalent non-aromatic condensed polycyclic group” as used herein may be a monovalent group (e.g., having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its molecular structure as a whole. Examples of a monovalent non-aromatic condensed polycyclic group may include an indenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, an indenophenanthrenyl group, an indenoanthracenyl group, and the like. The term “divalent non-aromatic condensed polycyclic group” as used herein may be a divalent group having a same structure as the monovalent non-aromatic condensed polycyclic group.
[0710] The term “monovalent non-aromatic condensed heteropolycyclic group” as used herein may be a monovalent group (e.g., having 1 to 60 carbon atoms) having two or more rings condensed to each other, further including, in addition to carbon atoms, at least one heteroatom as ring-forming atoms, and having no aromaticity in its molecular structure as a whole. Examples of a monovalent non-aromatic condensed heteropolycyclic group may include a pyrrolyl group, a thiophenyl group, a furanyl group, an indolyl group, a benzoindolyl group, a naphthoindolyl group, an isoindolyl group, a benzoisoindolyl group, a naphthoisoindolyl group, a benzosilolyl group, a benzothiophenyl group, a benzofuranyl group, a carbazolyl group, a dibenzosilolyl group, a dibenzothiophenyl group, a dibenzofuranyl group, an azacarbazolyl group, an azafluorenyl group, an azadibenzosilolyl group, an azadibenzothiophenyl group, an azadibenzofuranyl group, a pyrazolyl group, an imidazolyl group, a triazolyl group, a tetrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, an oxadiazolyl group, a thiadiazolyl group, a benzopyrazolyl group, a benzimidazolyl group, a benzoxazolyl group, a benzothiazolyl group, a benzoxadiazolyl group, a benzothiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an imidazotriazinyl group, an imidazopyrazinyl group, an imidazopyridazinyl group, an indenocarbazolyl group, an indolocarbazolyl group, a benzofurocarbazolyl group, a benzothienocarbazolyl group, a benzosilolocarbazolyl group, a benzoindolocarbazolyl group, a benzocarbazolyl group, a benzonaphthofuranyl group, a benzonaphthothiophenyl group, a benzonaphthosilolyl group, a benzofurodibenzofuranyl group, a benzofurodibenzothiophenyl group, a benzothienodibenzothiophenyl group, and the like. The term “divalent non-aromatic condensed heteropolycyclic group” as used herein may be a divalent group having a same structure as the monovalent non-aromatic condensed heteropolycyclic group.
[0711] The term “C.sub.6-C.sub.60 aryloxy group” as used herein may be a group represented by —O(A.sub.102)(wherein A.sub.102 may be a C.sub.6-C.sub.60 aryl group), and the term “C.sub.6-C.sub.60 arylthio group” as used herein may be a group represented by —S(A.sub.103)(wherein A.sub.103 may be a C.sub.6-C.sub.60 aryl group).
[0712] The term “C.sub.7-C.sub.60 arylalkyl group” as used herein may be a group represented by —(A.sub.104)(A.sub.105)(wherein A.sub.104 may be a C.sub.1-C.sub.54 alkylene group, and A.sub.105 may be a C.sub.6-C.sub.59 aryl group), and the term “C.sub.2-C.sub.60 heteroarylalkyl group” as used herein may be a group represented by —(A.sub.106)(A.sub.107)(wherein A.sub.106 may be a C.sub.1-C.sub.59 alkylene group, and A.sub.107 may be a C.sub.1-C.sub.59 heteroaryl group).
[0713] In the specification, the group “R.sub.10a” may be: [0714] deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; [0715] a C.sub.1-C.sub.60 alkyl group, a

C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or any combination thereof; [0716] a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, or a C.sub.2-C.sub.60 heteroarylalkyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or any combination thereof; or [0717] —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32).
[0718] In the specification, the groups Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof; a C.sub.7-C.sub.60 arylalkyl group; or a C.sub.2-C.sub.60 heteroarylalkyl group.
[0719] The term “heteroatom” as used herein may be any atom other than a carbon atom or a hydrogen atom. Examples of a heteroatom may include O, S, N, P, Si, B, Ge, Se, and any combination thereof.
[0720] In the specification, examples of a “third-row transition metal” may include hafnium (Hf), tantalum (Ta), tungsten (W), rhenium (Re), osmium (Os), iridium (Ir), platinum (Pt), gold (Au), and the like.
[0721] In the specification, the term “Ph” refers to a phenyl group, the term “Me” refers to a methyl group, the term “Et” refers to an ethyl group, the terms “ter-Bu” and “But” each refer to a tert-butyl group, and the term “OMe” refers to a methoxy group.
[0722] The term “biphenyl group” as used herein may be a “phenyl group substituted with a phenyl group.” For example, a “biphenyl group” may be a substituted phenyl group having a C.sub.6-C.sub.60 aryl group as a substituent.
[0723] The term “terphenyl group” as used herein may be a “phenyl group substituted with a biphenyl group.” For example, a “terphenyl group” may be a substituted phenyl group having, as a substituent, a C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group.
[0724] In the specification, the symbols * and *′, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula or moiety.
[0725] In the specification, the terms “x-axis”, “y-axis”, and “z-axis” are not limited to three axes in an orthogonal coordinate system (e.g., a Cartesian coordinate system), and may be interpreted in a broader sense than the aforementioned three axes in an orthogonal coordinate system. For example, the x-axis, y-axis, and z-axis may describe axes that are orthogonal to each other, or may describe axes that are in different directions that are not orthogonal to each other.
[0726] Hereinafter, compounds according to embodiments and light-emitting devices according to embodiments will be described in detail with reference to the Synthesis Examples and the

Examples. The wording "B was used instead of A" used in describing Synthesis Examples means that an identical molar equivalent of B was used in place of A.

Synthesis Examples and Examples

Synthesis Example 1: Synthesis of Compound 1

[0727] n-BuLi (1.6 M, 12.8 mL, 20.5 mmol) was added dropwise at −78° C. to a solution of 1,3-bis(2-(3-bromophenyl) propan-2-yl)benzene (4.72 g, 10.0 mmol) dissolved in anhydrous THF (100 mL). After the resultant solution was stirred for 1 hour, while stirring at −78° C., 10-phenylacridin-9 (10H)-one (2.98 g, 11.0 mmol) in a powder form was added thereto. The temperature was raised to room temperature, and the resultant reaction mixture was stirred for 3 hours. After quenching with a small amount of water, the resultant mixture was concentrated, and diluted with CH.sub.2Cl.sub.2 (300 mL). The resultant solution was washed with water, and an aqueous phase was extracted twice therefrom with CH.sub.2Cl.sub.2. A CH.sub.2Cl.sub.2 solution thus collected was dried with Na.sub.2SO.sub.4, and concentrated to obtain a hydroxy intermediate. The crude hydroxy intermediate was diluted with CH.sub.2Cl.sub.2 (200 mL), methanesulfonic acid (0.96 g, 10.0 mmol) was added thereto, and refluxed for 2 hours. After cooling to room temperature, the resultant solution was poured into an excess amount of aqueous NaHCO.sub.3 solution. The resultant product was subjected to an extraction process with CH.sub.2Cl.sub.2, and dried with Na.sub.2SO.sub.4. After filtration and evaporation, the resultant crude product was purified by silica gel column chromatography, and dried under reduced pressure to obtain Intermediate **1***a* in a solid form.

[0728] Boron triiodide (1.85 g, 4.7 mmol) was added to Intermediate **1***a* (2.44 g, 4.3 mmol) dissolved in anhydrous o-dichlorobenzene, and stirred at 140° C. for 24 hours. An excess amount of CH.sub.2Cl.sub.2 was added to the resultant mixture at room temperature, and stirred for 1 hour. After adding water, the resultant product was subjected to an extraction process with CH.sub.2Cl.sub.2. An organic layer thus collected was washed with brine, and dried with anhydrous Na.sub.2SO.sub.4. After filtration and evaporation, the resultant crude product was purified by silica gel column chromatography. After removing the solvent by evaporation, the resultant solid was washed with MeOH to obtain Compound 1 in a solid form (yield=0.69 g, 28%).

Synthesis Example 2: Synthesis of Compound 2

[0729] At 0° C. under N.sub.2, n-BuLi (1.6 M, 13.1 mL, 21.0 mmol) dissolved in hexane was added dropwise to a solution obtained by mixing and stirring 2-bromo-N-(2-g, bromophenyl)-N-phenylaniline (4.03 10.0 mmol) and 1,3-bis(2-(3-bromophenyl) propan-2-yl)benzene (4.72 g, 10.0 mmol) dissolved in anhydrous diethyl ether (100 mL). After the resultant mixture was stirred at 0° C. for 1 hour, tetrachlorosilane (1.90 g, 11.2 mmol) was added dropwise thereto. The resultant reaction mixture was stirred at 35° C. for 2 hours, and cooled to room temperature. After adding water, the resultant product was subjected to an extraction process with AcOEt. An organic layer thus collected was washed with brine, and dried with anhydrous Na.sub.2SO.sub.4. After filtration and evaporation, the resultant crude product was purified by silica gel column chromatography. After removing the solvent by evaporation, the resultant solid was washed with MeOH to obtain Intermediate **2***a* as a solid.

[0730] Boron triiodide (2.66 g, 6.8 mmol) was added to Intermediate **2***a* (3.62 g, 6.2 mmol) dissolved in anhydrous o-dichlorobenzene, and stirred at 140° C. for 24 hours. An excess amount of CH.sub.2Cl.sub.2 was added to the resultant mixture at room temperature, and stirred for 1 hour. After adding water, the resultant product was subjected to an extraction process with CH.sub.2Cl.sub.2. An organic layer thus collected was washed with brine, and dried with anhydrous Na.sub.2SO.sub.4. After filtration and evaporation, the resultant crude product was purified by silica gel column chromatography. After removing the solvent by evaporation, the resultant solid was washed with MeOH to obtain Compound 2 in a solid form (yield=2.97 g, 50%).

Synthesis Example 3: Synthesis of Compound 3

[0731] Compound 3 was synthesized in the same manner as in the synthesis of Compound 2, except that tetrachlorogermane (2.40 g, 11.2 mmol) was used instead of tetrachlorosilane (yield=2.16 g, 34%).

Synthesis Example 4: Synthesis of Compound 4

[0732] n-BuLi (1.6 M, 12.8 mL, 20.5 mmol) was added dropwise at −78° C. to a solution of 1,3-bis(9-(3-bromophenyl)-9H-fluoren-9-yl)benzene (7.17 g, 10.0 mmol) dissolved in anhydrous THF (100 mL). After the resultant solution was stirred for 1 hour, while stirring at −78° C., 10-([1,1′:3′,1″-terphenyl]-2′-yl) acridin-9 (10H)-one (4.66 g, 11.0 mmol) in a powder form was added thereto. The temperature was raised to room temperature, and the resultant reaction mixture was stirred for 3 hours. After quenching with a small amount of water, the resultant mixture was concentrated, and diluted with CH.sub.2Cl.sub.2 (300 mL). The resultant solution was washed with water, and an aqueous phase was extracted twice therefrom with CH.sub.2Cl.sub.2. A CH.sub.2Cl.sub.2 solution thus collected was dried with Na.sub.2SO.sub.4, and concentrated to obtain a hydroxy intermediate. The crude hydroxy intermediate was diluted with CH.sub.2Cl.sub.2 (200 mL), methanesulfonic acid (0.96 g, 10.0 mmol) was added thereto, and refluxed for 2 hours. After cooling to room temperature, the resultant solution was carefully poured into an excess amount of aqueous NaHCO.sub.3 solution. The resultant product was subjected to an extraction process with CH.sub.2Cl.sub.2, and dried with Na.sub.2SO.sub.4. After filtration and evaporation, the resultant crude product was purified by silica gel column chromatography, and dried under reduced pressure to obtain Intermediate **4***a* in a solid form.

[0733] Boron triiodide (1.25 g, 3.2 mmol) was added to Intermediate **4***a* (2.80 g, 2.9 mmol) dissolved in anhydrous o-dichlorobenzene, and stirred at 140° C. for 24 hours. An excess amount of CH.sub.2Cl.sub.2 was added to the resultant mixture at room temperature, and stirred for 1 hour. After adding water, the resultant product was subjected to an extraction process with CH.sub.2Cl.sub.2. An organic layer thus collected was washed with brine, and dried with anhydrous Na.sub.2SO.sub.4. After filtration and evaporation, the resultant crude product was purified by silica gel column chromatography. After removing the solvent by evaporation, the resultant solid was washed with MeOH to obtain Compound 4 in a solid form (yield=2.37 g, 24%).

Synthesis Example 5: Synthesis of Compound 5

[0734] Compound 5 was synthesized in the same manner as in the synthesis of Compound 2, except that N,N-bis(2-bromophenyl)-[1,1′: 3′,1″-terphenyl]-2′-amine (5.55 g, 10.0 mmol) was used instead of 2-bromo-N-(2-bromophenyl)-N-phenylaniline, and 1,3-bis(9-(3-bromophenyl)-9H-fluoren-9-yl)benzene (7.17 g, 10.0 mmol) was used instead of 1,3-bis(2-(3-bromophenyl) propan-2-yl)benzene (yield=3.06 g, 31%).

Synthesis Example 6: Synthesis of Compound 6

[0735] Compound 6 was synthesized in the same manner as in the synthesis of Compound 2, except that N,N-bis(2-bromophenyl)-[1,1′: 3′,1″-terphenyl]-2′-amine (5.55 g, 10.0 mmol) was used instead of 2-bromo-N-(2-bromophenyl)-N-phenylaniline, 1,3-bis(9-(3-bromophenyl)-9H-fluoren-9-yl)benzene (7.17 g, 10.0 mmol) was used instead of 1,3-bis(2-(3-bromophenyl) propan-2-yl)benzene, and tetrachlorogermane (2.40 g, 11.2 mmol) was used instead of tetrachlorosilane (yield=1.96 g, 19%).

[0736] 1H NMR and MS/FAB of the compounds synthesized according to Synthesis Examples are shown in Table 1. Synthesis methods of other compounds in addition to the compound synthesized in Synthesis Examples may be readily recognized by those skilled in the art by referring to the synthesis paths and source materials.

TABLE-US-00001 TABLE 1 Compound MS/FAB .sup.1H NMR Elemental Analysis No. found calc. found calc. 1 575.39 575.28 C, 89.75; H, 6.01; N, 2.48 C, 89.73; H, 5.95; N, 2.43 2 591.35 591.26 C, 85.31; H, 5.89; N, 2.40 C, 85.27; H, 5.79; N, 2.37 3 637.30 637.20 C, 79.35; H, 5.49; N, 2.21 C, 79.30; H, 5.39; N, 2.20 4 971.46 971.37 C, 92.70; H, 4.88; N, 1.46 C, 92.68; H, 4.77; N, 1.44 5 987.45 987.35 C, 89.98; H, 4.78; N, 1.43 C, 89.95; H, 4.69; N, 1.42 6 1033.41 1033.29 C, 86.10; H, 4.60; N, 1.37 C, 86.07; H, 4.49; N, 1.36

Evaluation Example 1

[0737] The HOMO energy level (E.sub.HOMO), LUMO energy level (E.sub.LUMO), HOMO-LUMO band gap energy (Eg), T.sub.1 energy level (nm), and existence ratio (%) of a triplet metal-to-ligand charge transfer state (.sup.3MLCT) of Compounds 1 to 6 were evaluated using the density functional theory (DFT) method of the Gaussian program, which was structure-optimized at the B3LYP/6-311G (d,p) level, and results thereof are shown in Table 2.

TABLE-US-00002 TABLE 2 Compound E.sub.HOMO E.sub.LUMO E.sub.g T.sub.1 No. (eV) (eV) (eV) (nm) 1 −5.14 −1.82 3.32 394 2 −5.28 −1.99 3.29 423 3 −5.25 −2.00 3.25 425 4 −5.17 −1.85 3.32 408 5 −5.31 −2.02 3.29 416 6 −5.29 −2.03 3.26 418

Example 1

[0738] As an anode, a glass substrate (product of Corning Inc.) with a 15 Ω/cm.sup.2 (1,200 Å) ITO formed thereon was cut to a size of 50 mm×50 mm×0.7 mm, sonicated by using isopropyl alcohol and pure water each for 5 minutes, washed by irradiation of ultraviolet rays and exposure of ozone thereto for 30 minutes, and mounted on a vacuum deposition apparatus.

[0739] 2-TNATA was vacuum-deposited on the anode to form a hole injection layer having a thickness of 600 Å. 4,4′-bis [N-(1-naphthyl)-N-phenylamino]biphenyl (hereinafter, referred to as NPB) was vacuum-deposited on the hole injection layer to form a hole transport layer having a thickness of 300 Å.

[0740] Compound 1, Compound DFD29, Compound HTH29, and Compound ETH2 were vacuum-deposited on the hole transport layer to form an emission layer having a thickness of 350 Å. The amount of Compound 1 was 13 parts by weight based on total 100 parts by weight of the emission layer, and the amount of Compound DFD29 was 0.4 parts by weight based on total 100 parts by weight of the emission layer. The balance of the emission layer by weight consisted of Compound HTH29 and Compound ETH2. The weight ratio of Compound HTH29 to Compound ETH2 was 6.5:3.5.

[0741] Compound HBL-1 was vacuum-deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å.

[0742] CNNPTRZ and LiQ were vacuum-deposited on the hole blocking layer to form an electron transport layer having a thickness of 310 Å. The weight ratio of CNNPTRZ to LiQ was 4:6.

[0743] Yb was vacuum-deposited on the electron transport layer to form an electron injection layer having a thickness of 15 Å.

[0744] Mg was vacuum-deposited on the electron injection layer to form a cathode having a thickness of 800 A, thereby completing the manufacture of an organic light-emitting device.

##STR00160## ##STR00161##

Examples 2 to 6 and Comparative Examples 1 to 3

[0745] Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compound used in forming the emission layer was changed as shown in Table 3.

Evaluation Example 1

[0746] The driving voltage (V) at 1,000 cd/m.sup.2, current efficiency (cd/A), maximum emission wavelength (nm), and lifespan (T95) of the organic light-emitting devices manufactured in Examples 1 to 6 and Comparative Examples 1 to 3 were measured using Keithley MU 236 and luminance meter PR650, and results thereof are shown in Table 3. The lifespan (T95) is a measure of the time (hr) taken until the luminance declines to 95% of the initial luminance.

[0747] In an embodiment, for the organic light-emitting devices manufactured in Examples 1 to 6 and Comparative Examples 1 to 3, the emission quantum yield (Φ.sub.PL) was determined through Quantaurus-QY, and the prompt (p) and delay (d) components were extracted using a streak camera to determine the exciton lifespan (T.sub.p, T.sub.d) and the luminescent component ratio (Φ.sub.p, Φ.sub.d). The RISC speed (K.sub.RISC) was calculated using the following calculation formulae, and results thereof are shown in Table 3.

[00001] $_{PL} = \ _p + \ _d \, k_r = \ _P / T_P \, k_{\mathrm{ISC}} = (1 - \ _P) / T_P$

$k_{RISC} = {}_{d} / (k_{ISC}$ .Math. T_p .Math. T_d .Math. ${}_{p})$

[0748] In Table 3, the RISC speed, current efficiency, and T95 show relative values with the values of Example 1 set to 100%.

TABLE-US-00003 TABLE 3 Driving Current Maximum Heterocyclic RISC voltage efficiency emission Lifespan compound speed (V) (cd/A) wavelength (nm) (T.sub.95, hr) Example 1 1 100% 4.0 100% 460 100% Example 2 2 190% 4.0 160% 460 120% Example 3 3 370% 4.0 220% 460 140% Example 4 4 100% 4.1 150% 460 240% Example 5 5 190% 4.1 230% 460 290% Example 6 6 370% 4.1 300% 460 340% Comparative CE1 90% 4.2 97% 460 21% Example 1 Comparative CE2 30% 5.0 65% 460 2% Example 2 Comparative CE3 80% 4.0 95% 460 13% Example 3 [00162]embedded image[00163]embedded image[00164]embedded image [00165]embedded image[00166]embedded image[00167]embedded image[00168] embedded image[00169]embedded image[00170]embedded image

[0749] Referring to Table 3, it was confirmed that the organic light-emitting devices according to Examples 1 to 6 emitted blue light and had superior RISC speed, luminescence efficiency, and device lifespan compared to the organic light-emitting devices according to Comparative Examples 1 to 3.

[0750] According to the embodiments, by using a heterocyclic compound, a light-emitting device having reduced driving voltage, improved color purity and efficiency, and increased lifespan, and a high-quality electronic apparatus including the light-emitting device may be manufactured.

[0751] Embodiments have been disclosed herein, and although terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purposes of limitation. In some instances, as would be apparent by one of ordinary skill in the art, features, characteristics, and/or elements described in connection with an embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of ordinary skill in the art that various changes in form and details may be made without departing from the spirit and scope of the disclosure

Claims

1. A light-emitting device comprising: a first electrode; a second electrode facing the first electrode; an interlayer between the first electrode and the second electrode and including an emission layer; and a heterocyclic compound including: an electron donor moiety including nitrogen; an electron acceptor moiety including boron; and a spiro core atom, wherein the spiro core atom is bonded to the electron donor moiety via a first bond and a second bond, the spiro core atom is bonded to the electron acceptor moiety via a third bond and a fourth bond, the first bond, the second bond, the third bond, and the fourth bond are each a single bond, and at least one of Conditions 1 to 3 is satisfied: [Condition 1] the electron donor moiety includes a group represented by Formula 1A; [Condition 2] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the first ring and the second ring are linked to each other via a first linking group; [Condition 3] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the second ring and the third ring are linked to each other via a second linking group; ##STR00171## wherein in Formula 1A, R.sub.61 to R.sub.63 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), a61 and a63 are each independently an integer from 1 to 5, a62 is an integer from 1 to 3, * indicates a binding site to a neighboring atom, R.sub.10a is: hydrogen, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.5-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or a combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or a combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or a combination thereof.

2. The light-emitting device of claim 1, wherein the emission layer includes a host and a dopant, and the dopant includes the heterocyclic compound.

3. The light-emitting device of claim 1, further comprising: a second compound including at least one π electron-deficient nitrogen-containing C.sub.1-C.sub.60 heterocyclic group; or a third compound including a group represented by Formula 3; or a fourth compound that emits delayed fluorescence; or a combination thereof, wherein the heterocyclic compound, the second compound, the third compound, and the fourth compound are different from each other; ##STR00172## wherein in Formula 3, ring CY.sub.71 and ring CY.sub.72 are each independently a Ir electron-rich C.sub.3-C.sub.60 cyclic group or a pyridine group, X.sub.71 is: a single bond; or a linking group including O, S, N, B, C, Si, or a combination thereof, and * indicates a binding site to any atom included in a remaining part of the third compound other than the group represented by Formula 3.

4. The light-emitting device of claim 3, wherein the second compound includes a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a combination thereof, and the fourth compound is a compound including at least one cyclic group including both boron (B) and nitrogen (N) as ring-forming atoms.

5. The light-emitting device of claim 3, wherein the emission layer includes: the heterocyclic compound; and the second compound, the third compound, the fourth compound, or a combination

thereof, and the emission layer emits blue light.
6. An electronic apparatus comprising the light-emitting device of claim 1.
7. The electronic apparatus of claim 6, further comprising: a thin-film transistor, wherein the thin-film transistor includes a source electrode and a drain electrode, and the first electrode of the light-emitting device is electrically connected to at least one of the source electrode and the drain electrode.
8. The electronic apparatus of claim 6, further comprising: a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or a combination thereof.
9. An electronic equipment comprising the light-emitting device of claim 1.
10. The electronic equipment of claim 9, wherein the electronic equipment is a flat panel display, a curved display, a computer monitor, a medical monitor, a television, a billboard, an indoor light, an outdoor light, a signal light, a head-up display, a fully transparent display, a partially transparent display, a flexible display, a rollable display, a foldable display, a stretchable display, a laser printer, a telephone, a mobile phone, a tablet computer, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality display, an augmented reality display, a vehicle, a video wall with multiple displays tiled together, a theater screen, a stadium screen, a phototherapy device, or a signboard.
11. A heterocyclic compound comprising: an electron donor moiety including nitrogen; an electron acceptor moiety including boron; and a spiro core atom, wherein the spiro core atom is bonded to the electron donor moiety via a first bond and a second bond, the spiro core atom is bonded to the electron acceptor moiety via a third bond and a fourth bond, the first bond, the second bond, the third bond, and the fourth bond are each a single bond, and at least one of Conditions 1 to 3 is satisfied: [Condition 1] the electron donor moiety includes a group represented by Formula 1A; [Condition 2] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the first ring and the second ring are linked to each other via a first linking group; [Condition 3] the electron acceptor moiety includes a first ring, a second ring, and a third ring each bonded to the boron, the first ring and the third ring are bonded to the spiro core atom, and the second ring and the third ring are linked to each other via a second linking group; ##STR00173## wherein in Formula 1A, R.sub.61 to R.sub.63 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), a61 and a63 are each independently an integer from 1 to 5, a62 is an integer from 1 to 3, * indicates a binding site to a neighboring atom, R.sub.10a is: hydrogen, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)

(Q.sub.12), —C(═O)(Q.sub.11), —S(═O).sub.2(Q.sub.11), —P(═O)(Q.sub.11)(Q.sub.12), or a combination thereof; a C.sub.5-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.5-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(═O)(Q.sub.21), —S(═O).sub.2(Q.sub.21), —P(═O)(Q.sub.21)(Q.sub.22), or a combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), or —P(═O)(Q.sub.31)(Q.sub.32), and Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or a combination thereof.

12. The heterocyclic compound of claim 11, wherein the heterocyclic compound is represented by Formula 1: ##STR00174## wherein in Formula 1, T.sub.1 is C, Si, or Ge, ring CY.sub.1 to ring CY.sub.5 are each independently a C.sub.5-C.sub.30 carbocyclic group or a C.sub.1-C.sub.30 heterocyclic group, T.sub.21 is *—C(Z.sub.1)(Z.sub.2)—**, *—Si(Z.sub.1)(Z.sub.2)—**, *—N(Z.sub.1)—**, *—O—**, or *—S—**, T.sub.22 is *—C(Z.sub.3)(Z.sub.4)—**, *—Si(Z.sub.3)(Z.sub.4)—**, *—N(Z.sub.3)—**, *—O—*′, or *—S—**, R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), a1 to a5 are each independently an integer from 1 to 20, Ar.sub.1 is a group represented by Formula 1A, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, Z.sub.1 and Z.sub.2 are optionally bonded to each other to form a C.sub.5-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, Z.sub.3 and Z.sub.4 are optionally bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, two or more neighboring groups among R.sub.1 in the number of a1, R.sub.2 in the number of a2, R.sub.3 in the number of a3, R.sub.4 in the number of a4, R.sub.5 in the number of a5, Z.sub.1 to Z.sub.4, and Ar.sub.1 are optionally bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and R.sub.10a and Q.sub.1 to Q.sub.3 are each the same as defined in

Formula 1A.

13. The heterocyclic compound of claim 12, wherein ring CY.sub.1 to ring CY.sub.5 are each independently a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an azafluorene group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a dibenzooxasiline group, a dibenzothiasiline group, a a dibenzodihydroazasiline group, dibenzodihydrodihydrodisiline group, a dibenzodihydrosiline group, a dibenzodioxine group, a dibenzooxathiine group, a dibenzooxazine group, a dibenzopyran group, a dibenzodithiine group, a dibenzothiazine group, a dibenzothiopyran group, a dibenzocyclohexadiene group, a dibenzodihydropyridine group, or a dibenzodihydropyrazine group.

14. The heterocyclic compound of claim 12, wherein T.sub.21 is *—C(Z.sub.1)(Z.sub.2)—** or *—Si(Z.sub.1)(Z.sub.2)—**, and T.sub.22 is *—C(Z.sub.3)(Z.sub.4)—** or *—Si(Z.sub.3)(Z.sub.4)—**.

15. The heterocyclic compound of claim 14, wherein at least one of Conditions 4 and 5 is satisfied: [Condition 4] Z.sub.1 and Z.sub.2 are bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a; [Condition 5] Z.sub.3 and Z.sub.4 are bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

16. The heterocyclic compound of claim 12, wherein R.sub.1 to R.sub.5 and Z.sub.1 to Z.sub.4 are each independently: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.10 alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, or an azadibenzosilolyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or a combination thereof; or —C(Q.sub.1)(Q.sub.2)(Q.sub.3), —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(═O)(Q.sub.1), —S(═O).sub.2(Q.sub.1), or —P(═O)(Q.sub.1)(Q.sub.2), and Q.sub.1 to Q.sub.3 and Q.sub.31 to Q.sub.33 are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or a combination thereof.

17. The heterocyclic compound of claim 12, wherein Ar.sub.1 is: a group represented by Formula 1A; or a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azafluorenyl group, or an azadibenzosilolyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —

CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a biphenyl group, a C.sub.1-C.sub.10 alkylphenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —O(Q.sub.31), —S(Q.sub.31), —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —P(Q.sub.31)(Q.sub.32), —C(═O)(Q.sub.31), —S(═O).sub.2(Q.sub.31), —P(═O)(Q.sub.31)(Q.sub.32), or a combination thereof, and Q.sub.31 to Q.sub.33 are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or a combination thereof.

18. The heterocyclic compound of claim 12, wherein Ar.sub.1 is a group represented by Formula 1A.

19. The heterocyclic compound of claim 12, wherein T.sub.21 and T.sub.22 are each independently a group represented by Formula 1B: ##STR00175## wherein in Formula 1B, R.sub.10a is the same as defined in Formula 1A, b1 and b2 are each independently an integer from 1 to 4, T.sub.3 is a single bond, *—O—**, *—S—**, *—N(R′)—**, *—C(R′)(R″)—**, or *—Si(R′)(R″)—**, R′ and R″ are each independently the same as defined in connection with Q.sub.1 in Formula 1A, and * and *′ each independently indicate a binding site to a neighboring atom.

20. The heterocyclic compound of claim 12, wherein the heterocyclic compound is represented by Formula 1-1: ##STR00176## wherein in Formula 1-1, T.sub.1, T.sub.21, T.sub.22, and Ar.sub.1 are each the same as defined in Formula 1, X.sub.11 is C(R.sub.11) or N, X.sub.12 is C(R.sub.12) or N, X.sub.13 is C(R.sub.13) or N, X.sub.14 is C(R.sub.14) or N, X.sub.21 is C(R.sub.21) or N, X.sub.22 is C(R.sub.22) or N, X.sub.23 is C(R.sub.23) or N, X.sub.31 is C(R.sub.31) or N, X.sub.32 is C(R.sub.32) or N, X.sub.33 is C(R.sub.33) or N, X.sub.41 is C(R.sub.41) or N, X.sub.42 is C(R.sub.42) or N, X.sub.43 is C(R.sub.43) or N, X.sub.51 is C(R.sub.51) or N, X.sub.52 is C(R.sub.52) or N, X.sub.53 is C(R.sub.53) or N, X.sub.54 is C(R.sub.54) or N, R.sub.11 to R.sub.14 are each independently the same as defined in connection with R.sub.1 in Formula 1, R.sub.21 to R.sub.23 are each independently the same as defined in connection with R.sub.2 in Formula 1, R.sub.31 to R.sub.33 are each independently the same as defined in connection with R.sub.3 in Formula 1, R.sub.41 to R.sub.43 are each independently the same as defined in connection with R.sub.4 in Formula 1, R.sub.51 to R.sub.54 are each independently the same as defined in connection with R.sub.5 in Formula 1, and two or more neighboring groups among R.sub.11 to R.sub.14, R.sub.21 to R.sub.23, R.sub.31 to R.sub.33, R.sub.41 to R.sub.43, R.sub.51 to R.sub.54, Z.sub.1 to Z.sub.4, and Ar.sub.1 are optionally bonded to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.